

Reflected max-convolution inequalities for finite supports

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Abstract

For every nonnegative finitely supported kernel k on the integers, write $K = \|k\|_1$ and $M = \|k\|_\infty$. For all nonzero finitely supported $f, g \geq 0$, we prove

$$\frac{\|f \star \widetilde{gk}\|_1}{\|f \star g\|_1} \leq \frac{(\sqrt{K} + \sqrt{M})^2}{4}.$$

The proof uses the mixed weighted form of the classical equality of sum and difference representation energies. A weighted allocation argument also gives the bound $\max\{K/2, (K + 3M)/4\}$. For $k = 1_F$, $|F| = n$, the bound is $(\sqrt{n} + 1)^2/4$, strict for each input pair when $n > 1$. A geometric product construction gives ratios tending to $(K + 3M)/4$ for prescribed kernel values with varying integer locations. In particular, the supremum on n sites is at least $(n + 3)/4$ for $n \geq 2$, and the leading coefficient $1/4$ is asymptotically optimal. For every prescribed value list with $K \leq 3M$ or at most four entries, the supremum is exactly $(K + 3M)/4$. The support suprema on one through four sites are therefore 1 , $5/4$, $3/2$ and $7/4$, respectively.

The four-site upper bound uses an inequality for four events and finite linear-programming duality. Averaging over four-subsets gives the bounds $35/16$ and $21/8$ on five and six sites. We deduce the strict reflected fourfold inequality for constant kernels on every five-point support and on six-point supports with at least 48 fourfold sums. For constant kernels it also holds on every B_4 set; for nonconstant kernels on B_4 supports it holds when $K/M \geq 9$. Here B_4 means that unordered four-term sums are distinct, allowing repetition.

For kernels on at most four points, we prove a stronger covering bound by finite linear-programming duality and classify the five pair-sum collision patterns, obtaining their sharp unweighted covering constants. The Sidon set $\{0, 2, 7, 8, 11\}$ violates that covering bound by $1/16$, establishing its support-size limit. On this support the reflected fourfold inequality nevertheless holds for arbitrary nonnegative kernel weights. Its proof uses an allocation estimate and 240 rational coefficient certificates covering every weight order, with strictness for positive kernel weights and nonzero inputs.

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*Developed with AI research agents. The sharp five-point Sidon cardinality bound uses a cited classical small-sumset theorem; the arbitrary-weight five-point result uses rational certificates checked by the supplied program.

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1 The inequality and its covering constant

All functions below are nonnegative and finitely supported on $\mathbb{Z}$. We distinguish ordinary convolution from max-convolution:

$$(u * v)(s) = \sum_{p+q=s} u(p)v(q), \quad (u \star v)(s) = \max_{p+q=s} u(p)v(q).$$

The maximum is zero when no pair contributes. We write $\tilde{u}(p) = u(-p)$ and $\|u\|_1 = \sum_p u(p)$. For a finite nonempty set F and a kernel $k \geq 0$ supported on F , define

$$N_F(k) = \max_{x \geq 0, \supp x \subseteq F} \sum_{p \in F} k(p)x(p)^2, \quad T_F(k) = \|k \star k \star k \star k\|_1. \quad (1.1)$$

The constraint is coefficientwise. Each coordinate of x is at most one, so the maximum exists by compactness.

Theorem 1.1. *If $1 \leq |F| \leq 4$, then every nonnegative kernel supported on F satisfies*

$$N_F(k)^4 \leq T_F(k). \quad (1.2)$$

If $|F| = 4$ and k is positive at every point of F , the inequality is strict. For $F = \{0, 2, 7, 8, 11\}$ and $k = 1_F$, instead

$$N_F(k)^4 \geq (5/2)^4 = 39 + 1/16 > 39 = T_F(k).$$

Thus four is the largest support size for which (1.2) holds universally.

The theorem concerns a covering bound for the reflected max-convolution ratio. The connection follows from a general finite-dimensional lemma.

Lemma 1.2 (Universal translated covering). *For $g \geq 0$, nonzero and supported on F , define*

$$L_F(g, k) = \min_{\lambda_t \geq 0} \left\{ \sum_{t \in F+F} \lambda_t : \sum_{q \in F} g(q)\lambda_{p+q} \geq g(p)k(p) \quad (p \in F) \right\}. \quad (1.3)$$

Then

$$\sup_{g \neq 0, g \geq 0} L_F(g, k) = N_F(k). \quad (1.4)$$

The same supremum results if g is required to be positive throughout F . For every nonzero finitely supported $f \geq 0$,

$$\|f \star \widetilde{gk}\|_1 \leq L_F(g, k)\|f \star g\|_1 \leq N_F(k)\|f \star g\|_1. \quad (1.5)$$

Here gk denotes pointwise multiplication.

Proof. Fix $q_0 \in F$ with $g(q_0) > 0$. Set $\lambda_{p+q_0} = g(p)k(p)/g(q_0)$ for $p \in F$, and set all remaining variables to zero. The shifts $p + q_0$ are distinct, so this is feasible. A bounded minimizing level set is compact. The finite linear-programming dual maximizes $\sum_p k(p)g(p)z(p)$ subject to $z \geq 0$ and $z * g \leq 1$. It contains zero and is bounded, since $z(p) \leq 1/g(q_0)$. Both extrema are attained and are equal by finite linear-programming duality.

For a feasible dual vector set $x(p) = \sqrt{g(p)z(p)}$. Coefficientwise,

$$(x * x)(s) \leq \frac{1}{2} \sum_{p+q=s} (g(p)z(q) + g(q)z(p)) = (z * g)(s) \leq 1.$$

Thus $L_F(g, k) \leq N_F(k)$. Conversely, for a nonzero maximizer x in (1.1), taking $g = z = x$ attains its objective in the dual. If $k \neq 0$, homogeneity ensures $\|x * x\|_\infty = 1$ at a maximizer. With $g_\epsilon = x + \epsilon 1_F$ and $z_\epsilon = g_\epsilon / \|g_\epsilon * g_\epsilon\|_\infty$, the dual value tends to $N_F(k)$. This proves the positive- g assertion; $k = 0$ is immediate.

An attained cover gives, with $h = \widetilde{gk}$,

$$h(v) \leq \sum_t \lambda_t g(v+t), \quad (f \star h)(u) \leq \sum_t \lambda_t (f \star g)(u+t).$$

The first inequality is the constraint at $p = -v$ when $v \in -F$ and is automatic elsewhere. Summing the second over u proves (1.5). $\square$

Corollary 1.3. *If $|F| \leq 4$, $g \geq 0$ is nonzero and supported on F , and $k \geq 0$ is supported on F , then every nonzero finitely supported $f \geq 0$ satisfies*

$$\|f \star \widetilde{gk}\|_1^4 \leq \|f \star g\|_1^4 \|k \star k \star k \star k\|_1.$$

The inequality is strict when $|F| = 4$ and $k > 0$ on F , including when some values of g are zero.

Proof. Apply Theorem 1.1 to Lemma 1.2; $\|f \star g\|_1 > 0$. $\square$

Theorem 11.1 proves the same reflected inequality for arbitrary kernels on $F = \{0, 2, 7, 8, 11\}$, where the stronger bound (1.2) fails. Its proof uses the actual common input f, g to improve the covering estimate before comparing with $T_F(k)$.

Translation and nonzero dilation of F relabel pair sums and fourfold sums injectively and preserve N_F and T_F . The translated covering argument uses shifts in the actual integer support, so it imposes no residue-class condition on f . Reflection is included. Multiplying k by a constant scales N_F linearly and T_F to the fourth power.

1.1 Relation to sumset inequalities

Gyarmati, Hennecart and Ruzsa [1, p. 177, equation (9)] asked when

$$|A - B| \leq |A + B| |hB|^{1/h}$$

holds for all finite nonempty sets in an abelian group, where hB is the repeated sumset. Their paper gives a sixfold counterexample and leaves four and five unresolved [1, p. 182]. With indicator functions and $k = 1_F$, Corollary 1.3 and Theorem 10.14 give the fourfold inequality for $B = F$ of size at most five. Both allow arbitrary f and g ; the four-point result also permits arbitrary kernel weights k . The five-point example in Theorem 1.1 concerns N_F and satisfies the reflected inequality for every kernel by Theorem 11.1. Corollary 10.13 also proves the constant-kernel fourfold inequality on six-point supports with $|4F| \geq 48$. Corollary 10.20 also gives the constant-kernel inequality on B_4 sets of every finite size. These

are sets with distinct unordered four-term sums, including repetitions. Corollary 10.21 also permits nonconstant kernels on these supports when $\|k\|_1/\|k\|_\infty \geq 9$.

Max-convolution and weighted sumset parameters are established tools. Matolcsi, Ruzsa, Shakan and Zhelezov [6, Definition 8.1] use the operation above and relate a weighted induced-tripling parameter to a set parameter in their Theorem 8.6. The cube norm inequality of Becker, Ivanisvili, Krachun and Madrid [7, Theorem 1.3] and the mixed-alphabet bounds of Hosle and Ivanisvili [8, Theorem 1.5] concern lower bounds for max-convolution and sumset expansion. Green, Matolcsi, Ruzsa, Shakan and Zhelezov [12, Proposition 2.1] prove the weighted discrete Prékopa–Leindler inequality

$$\sum_n \max\{p(a \star b)(n), (1-p)(a \star b)(n-1)\} \geq \|a\|_2 \|b\|_2 \quad (0 \leq p \leq 1),$$

and deduce a triple-sum lower bound for subsets of quasicubes in their Theorem 1.1. Their proposition is equivalent to the $p = q = 2$ case of Theorem 11.1 of Matolcsi et al. [6]. For ordinary autoconvolution, Cloninger and Steinerberger [9, Section 1.2] and Ivanisvili and Xie [10, Theorem 5] study continuous L^1 mass on a fixed interval. That normalization differs from the weighted squared input mass in $N_F(k)$ and its comparison with $T_F(k)$.

These differences in theorem statements establish neither novelty nor non-implication. The covering proofs and the constant-kernel five-point fourfold theorem are self-contained. The stronger Sidon cardinality bound $|4F| \geq 39$ uses Freiman’s classical $3k-4$ theorem, in the form recalled by Bollobás, Leader and Tiba [11, p. 1]. Priority in the wider literature remains unestablished. In particular, the mathematical comparison with the weighted Prékopa–Leindler results still requires control of the linked ratio involving $f \star g$ and $f \star gk$, rather than a lower bound for an independent max-convolution. The companion [literature comparison](#) records the exact source versions, reading limits and unreproduced numerical dependencies of the continuous results.

1.2 A coefficient comparison used below

Let $v \geq 0$ be a profile on F . Group the expansion of $(k \cdot v)^4$ by the sum of its four support positions. If the total coefficient weight in each group is at most one, then

$$(k \cdot v)^4 \leq T_F(k). \tag{1.6}$$

Indeed, each kernel monomial in that group is at most its max-convolution value. The same argument applies after any stated nonnegative AM–GM replacement of monomials. Such a replacement may move weight to another sum; the proof must account for the coefficient total at every destination. A strict deficit at a group with positive maximum proves strictness. A profile used in (1.6) need not itself be feasible for the ordinary-autoconvolution constraint.

2 The five pair-sum patterns

The ordinary-convolution constraints depend only on equalities between unordered pair sums, with repeated summands allowed. For four integer points these equalities give five patterns.

Proposition 2.1. *Every four-point integer set belongs to exactly one of the following classes: a Sidon set, an isolated three-term arithmetic progression, a proper parallelogram, two three-term arithmetic progressions, or a four-term arithmetic progression. Here a Sidon set has distinct unordered pair sums. An isolated progression has exactly one nontrivial pair-sum*

equality, of the form $p + r = 2q$. A proper parallelogram has exactly one nontrivial pair-sum equality, with four distinct summands.

Up to translation, reflection, and integer dilation, the class with two progressions is represented by $\{0, 1, 2, 4\}$.

Proof. Translate the set to $\{0, a, b, c\}$, where $0 < a < b < c$. A shared summand can be cancelled from any pair-sum equality. The remaining nontrivial possibilities are precisely

$$b = 2a, \quad c = 2a, \quad c = 2b, \quad c = 2b - a, \quad c = a + b. \quad (2.1)$$

The first four assert that one point is the midpoint of two others. The last is the only possible equality between pairs using all four points.

The compatible intersections are as follows. The equations $b = 2a$ and $c = 2b$ give $\{0, a, 2a, 4a\}$. The equations $c = 2a$ and $c = 2b - a$ give $b = 3a/2$, hence a reflected copy of $\{0, d, 2d, 4d\}$ with $d = a/2$. Any two of $b = 2a$, $c = 2b - a$, and $c = a + b$ imply the third and give $\{0, a, 2a, 3a\}$. Every other pair of equations in (2.1) contradicts $0 < a < b < c$. The possibilities are therefore no equality, one midpoint equality, one cross-pair equality, the two-progressions case, and the four-term progression. $\square$

This classification concerns pair sums. A class can contain further equalities between fourfold sums, which must also be accounted for when bounding $T_F(k)$.

3 Sidon supports

Lemma 3.1. *For every finitely supported nonnegative function h on $\mathbb{Z}$,*

$$\|h \star h\|_1 \geq 2 \sum_s h(s)^2 - \|h\|_\infty^2. \quad (3.1)$$

Proof. The zero function is immediate. For $0 \leq t < \|h\|_\infty^2$, put $A_t = \{s : h(s)^2 > t\}$. Then A_t is nonempty and $A_t + A_t \subseteq \{s : (h \star h)(s) > t\}$. If a finite integer set has ordered elements $a_1 < \dots < a_m$, the chain

$$2a_1 < a_1 + a_2 < \dots < a_1 + a_m < a_2 + a_m < \dots < 2a_m$$

shows that $|A + A| \geq 2|A| - 1$. Integrating the resulting inequality for superlevel-set cardinalities over $0 \leq t < \|h\|_\infty^2$ proves (3.1). $\square$

Proposition 3.2. *Let F be a finite nonempty Sidon set and let $k \geq 0$ on F . Put $K = \sum_{p \in F} k(p)$ and $a = \max_{p \in F} k(p)$. Then*

$$N_F(k) = \max \left\{ \frac{K}{2}, \frac{K + 3a}{4} \right\}. \quad (3.2)$$

If $|F| \leq 4$, then $N_F(k)^4 \leq T_F(k)$, with strict inequality whenever at least two values of k are positive.

Proof. The Sidon property makes the ordinary-convolution constraints equivalent to

$$w_p \leq 1, \quad w_p w_q \leq \frac{1}{4} \quad (p \neq q), \quad w_p = x(p)^2 \geq 0.$$

If all $w_p \leq 1/2$, the objective is at most $K/2$. Otherwise let $w_j = u \in [1/2, 1]$ be largest. Every other coordinate is at most $1/(4u)$, so

$$\sum_p k(p) w_p \leq k(j) u + \frac{K - k(j)}{4u}.$$

This is convex in u , and its endpoint values are $K/2$ and $k(j) + (K - k(j))/4$. Maximizing over j proves the upper bound in (3.2). The profiles with all coordinates $1/2$, and with one coordinate 1 and all others $1/4$, are feasible. They attain the stated values.

Suppose now that $|F| \leq 4$. The case $k = 0$ is immediate. Apply Lemma 3.1 to $h = k \star k$. Distinct unordered pair sums give

$$\sum_s h(s)^2 = \frac{1}{2} \left[\left(\sum_p k(p)^2 \right)^2 + \sum_p k(p)^4 \right], \quad \|h\|_\infty = a^2.$$

Associativity of max-convolution and Cauchy–Schwarz therefore imply

$$T_F(k) \geq \left(\sum_p k(p)^2 \right)^2 + \sum_p k(p)^4 - a^4 \geq \left(\frac{K}{2} \right)^4. \quad (3.3)$$

The last comparison is strict if at least two kernel values are positive, because then $\sum_p k(p)^4 - a^4 > 0$.

For the other term in (3.2), choose a point p_0 with $k(p_0) = a$ and write $R = K - a$. Fixing two copies of p_0 in the fourfold maximum gives distinct sums indexed by unordered pairs from F . Thus

$$T_F(k) \geq a^4 + a^3 R + a^2 h_2 \geq a^4 + a^3 R + \frac{2}{3} a^2 R^2, \quad (3.4)$$

where h_2 is the sum of the squares and pair products of the other kernel values. There are at most three such values, so $h_2 = (R^2 + \sum_{p \neq p_0} k(p)^2)/2 \geq 2R^2/3$. Since $R \leq 3a$, expansion gives

$$\left(a + \frac{R}{4} \right)^4 \leq a^4 + a^3 R + \frac{153}{256} a^2 R^2. \quad (3.5)$$

Indeed the coefficient after bounding the cubic and quartic terms is $3/8 + 3/16 + 9/256 = 153/256$. The difference between the right sides of (3.4) and (3.5) is at least $53a^2 R^2/768$, positive when $R > 0$. This proves both comparisons in (3.2), including the asserted strictness. $\square$

4 Proper parallelograms

Write

$$F = \{0, a, b, a + b\}, \quad 0 < a < b, \quad b \neq 2a, \quad (4.1)$$

and index its coordinates in this order. The opposite pairs are $\{0, 3\}$ and $\{1, 2\}$. Define a collection $\mathcal{P}$ of sixteen squared profiles:

- Four profiles have three coordinates $1/2$ and one coordinate zero.
- Eight profiles have one coordinate 1 , its opposite coordinate and one adjacent coordinate $1/4$, and the remaining coordinate zero.
- Four profiles have one coordinate 1 , its two adjacent coordinates $1/4$, and its opposite coordinate $1/16$.

We call the last four profiles tensor profiles, since their entries are products of the two-coordinate vector $(1, 1/4)$.

Proposition 4.1. *For every $k \geq 0$ on (4.1),*

$$N_F(k) = \max_{v \in \mathcal{P}} k \cdot v, \quad (k \cdot v)^4 \leq T_F(k) \quad (v \in \mathcal{P}). \quad (4.2)$$

If k is positive at all four points, every displayed quartic comparison is strict, and hence $N_F(k)^4 < T_F(k)$.

4.1 Reduction to sixteen profiles

Ordinary-convolution feasibility is equivalent to

$$x_i^2 \leq 1, \quad 2x_i x_j \leq 1 \quad (i \neq j), \quad x_0 x_3 + x_1 x_2 \leq \frac{1}{2}. \quad (4.3)$$

The last constraint is the coefficient at $a+b$; its separate opposite-pair bounds are redundant. Each profile in $\mathcal{P}$ is feasible. For the first two types the sum of the two opposite products of square roots is $1/2$; for a tensor profile both products are $1/4$.

Put $w_i = x_i^2$. First suppose every $w_i \leq 1/2$. With $r = x_0 x_3$ and $s = x_1 x_2$, the inequality $u + v \leq 1/2 + 2uv$ for $u, v \in [0, 1/2]$ gives

$$\sum_i w_i \leq 1 + 2r^2 + 2s^2 \leq 1 + 2(r + s)^2 \leq \frac{3}{2}.$$

Increase coordinates within $[0, 1/2]$ until their sum is $3/2$. The resulting vector lies in the convex hull of the four profiles of the first type: for such a vector y , the coefficient of the profile with its i th coordinate zero is $1 - 2y_i$. These coefficients are nonnegative and sum to one.

Otherwise relabel corners, preserving opposite pairs, so that $w_0 = u \geq 1/2$, and put $L = 1/(4u)$. Then $w_1, w_2, w_3 \leq L$, and

$$w_3 \leq L(1 - 2\sqrt{w_1 w_2})^2.$$

For fixed u , the upper bound

$$k_0 u + k_1 w_1 + k_2 w_2 + k_3 L(1 - 2\sqrt{w_1 w_2})^2$$

is separately convex in $w_1, w_2 \in [0, L]$. Its only nonlinear term in either variable is a non-positive multiple of that variable's square root. It is therefore enough to consider the four corners of this square.

The corner $(0, 0)$ is dominated by either $(L, 0)$ or $(0, L)$. These give $(u, L, 0, L)$ and $(u, 0, L, L)$, whose objectives are convex in u . Their endpoint values correspond to profiles of the first two types. The last corner gives

$$b(u) = \left(u, L, L, \frac{(2u - 1)^2}{16u^3} \right).$$

For this curve, set

$$t = 2u - 1, \quad \nu = 1 - t, \quad s = \frac{(2u - 1)(1 - u)}{4u},$$

and define

$$\begin{aligned} U &= (1, 1/4, 1/4, 1/16), & W_0 &= (1/2, 1/2, 1/2, 0), \\ W_1 &= (1/2, 1/2, 0, 1/2), & W_2 &= (1/2, 0, 1/2, 1/2). \end{aligned}$$

The coefficients of

$$y = tU + (\nu - 4s)W_0 + 2sW_1 + 2sW_2$$

are nonnegative and sum to one, since $4s = t(1 - u)/u \leq 2(1 - u) = \nu$. The first three coordinates of y are (u, L, L) and its last coordinate is $t/16 + 2s$. Moreover,

$$\frac{(2u - 1)^2}{16u^3} - \frac{t}{16} = \frac{t(1 - u)(u^2 + u - 1)}{16u^3} \leq \frac{s}{4} \leq 2s.$$

Thus y dominates $b(u)$. These reductions prove the upper bound in (4.2); feasibility of the listed profiles proves equality.

4.2 Profiles supported on three corners

Every three-corner subset of a proper parallelogram is Sidon. An additional midpoint equality would force $b = 2a$ or $a = b$. For a kernel restricted to such a triangle, Proposition 3.2 bounds both its half-sum and every unit-and-two-quarter form. These are exactly the objectives of the twelve profiles with a zero coordinate. Monotonicity under enlarging the support then proves their quartic comparisons with $T_F(k)$. For positive k , their three-point restrictions have at least two positive values, so the comparisons are strict.

4.3 Tensor profiles and their collision grid

Reflection around $(a + b)/2$ reduces the four tensor profiles to a pivot at 0 and a pivot at a . Relative to the pivot, write the kernel values as A, B, C, D , with the tensor objective

$$E = A + B/4 + C/4 + D/16.$$

For the outer pivot the offsets are $0, a, b, a + b$. For the inner pivot they are $0, -a, b, b - a$, so that $A = k(a)$, $B = k(0)$, $C = k(a + b)$, and $D = k(b)$ in that case.

A monomial with multiplicities n_A, n_B, n_C, n_D has grid coordinates $i = n_B + n_D$ and $j = n_C + n_D$. Both belong to $\{0, 1, 2, 3, 4\}$. The total coefficient of its grid point in E^4 is

$$c_i c_j, \quad (c_0, c_1, c_2, c_3, c_4) = (1, 1, 3/8, 1/16, 1/256), \quad (4.4)$$

because the coefficient-generating polynomial is

$$(1 + z/4 + w/4 + zw/16)^4 = (1 + z/4)^4(1 + w/4)^4.$$

The actual sum is $ia + jb$ for the outer pivot and $-ia + jb$ for the inner pivot. Grouping by that sum proves $E^4 \leq T_F(k)$ whenever every group's nonnegative coefficient weights add to at most one. AM–GM inequalities can first move terms between groups, provided each replacement is a legal fourfold monomial at its new sum.

Write $b/a = r/s$ in lowest terms, with $r > s > 0$. Two grid points in $\{0, \dots, 4\}^2$ collide only if $r \leq 4$. Excluding $r/s = 2$ leaves

$$r/s \in \{3, 4, 3/2, 4/3\}. \quad (4.5)$$

The collision step is $(r, -s)$ for the outer pivot and (r, s) for the inner pivot. Since $r \geq 3$, no group contains three grid points. Outside (4.5), (4.4) proves the quartic comparison directly.

We use two elementary inequalities:

$$U + \epsilon V \leq \max(U, V) + \epsilon \min(U, V) \quad (U, V \geq 0, 0 \leq \epsilon \leq 1), \quad (4.6)$$

$$\min(A^3 C, X^4) \leq A^2 X^2 + A^2 C^2 \quad (A, C, X \geq 0). \quad (4.7)$$

The first follows by ordering U, V . For the second, the case $A = 0$ is immediate. Otherwise divide by A^4 and put $x = X/A$, $y = C/A$. If $x \leq 1$, use $\min(y, x^4) \leq x^4 \leq x^2$; if $x \geq 1$ and $y \leq 1$, use $y \leq x^2$; if $y \geq 1$, use $y \leq y^2$.

4.4 Outer-pivot collisions

At ratios $3/2$ and $4/3$, every colliding pair has total weight at most $3/8 + 1/16 = 7/16$. Indeed the point with smaller first coordinate has second coordinate at least two, and the other point has first coordinate at least three. Thus the coefficient comparison applies directly.

At ratio three, the only groups of weight greater than one are $(0, 1) \sim (3, 0)$ and $(1, 1) \sim (4, 0)$. Their polynomials are

$$A^3C + AB^3/16, \quad 3A^2BC/4 + A^3D/4 + B^4/256.$$

Use $AB^3/16 \leq (A^2B^2 + B^4)/32$ in the first group. The resulting total coefficient of B^4 in the second group is $\delta = 9/256$. Keep

$$(3/4 - \delta)A^2BC + A^3D/4 + \delta B^4$$

there; its coefficient sum is one. The displaced term satisfies $\delta A^2BC \leq \delta(A^2B^2 + A^2C^2)/2$. The extra coefficients assigned to A^2B^2 and A^2C^2 are $25/512$ and $9/512$. Their entire original group weights are $3/8$ and $7/16$, respectively, so the final weights are $217/512$ and $233/512$. Both are below one.

At ratio four, the sole overloaded group is $A^3C + B^4/256$. Equations (4.6)–(4.7), with $\epsilon = 1/256$ and $X = B$, bound it by its maximum plus $(A^2B^2 + A^2C^2)/256$. The two recipient groups originally have weights $3/8$ and $97/256$; their final weights are $97/256$ and $98/256$. Every other group already has weight at most one.

4.5 Inner-pivot collisions

An inner-pivot group can be overloaded only if the smaller grid point has both coordinates at most one. Otherwise its weight is at most $3/8$, and that of its partner is at most $1/16$. For ratios three, four, and $3/2$, remove the high-point polynomials in the following table and keep their low-point partners in full.

High grid point	Polynomial
(3, 1)	$B^3C/64 + 3AB^2D/64$
(4, 1)	$B^3D/256$
(3, 2)	$3B^2CD/256 + 3ABD^2/256$
(4, 2)	$3B^2D^2/2048$
(3, 3)	$3BCD^2/1024 + AD^3/1024$
(4, 3)	$BD^3/4096$

The entries follow from the multinomial expansion of E^4 . The sets of removed points and their total coefficient weights are

Ratio	Removed points	Total weight
3	(3, 1), (4, 1), (3, 2), (4, 2)	$187/2048$
4	(4, 1), (4, 2)	$11/2048$
$3/2$	(3, 2), (4, 2), (3, 3), (4, 3)	$119/4096$

Each total is below $1/8$. Every unweighted monomial in the first table is at most

$$S = B^4 + C^4 + D^4 + A^2B^2 + A^2D^2.$$

For monomials without A , this follows from weighted AM–GM. The three remaining forms satisfy

$$AB^2D \leq \frac{A^2D^2 + B^4}{2}, \quad ABD^2 \leq \frac{A^2B^2 + D^4}{2}, \quad AD^3 \leq \frac{A^2D^2 + D^4}{2}.$$

The removed polynomial is consequently bounded by $S/8$.

The five recipient monomials occupy distinct groups, outside the removed groups. In the order $B^4, C^4, D^4, A^2B^2, A^2D^2$, their original weights are

$$\frac{1}{256}, \quad \frac{1}{256}, \quad \begin{cases} 1/16 + 1/65536, & r/s = 3 \text{ or } 4, \\ 3/8 + 1/65536, & r/s = 3/2, \end{cases} \quad \frac{3}{8}, \quad \frac{9}{64}.$$

These are full group weights, including any collision partner. For example, the partner of the D^4 point $(4, 4)$ is $(1, 3)$, $(0, 3)$, or $(1, 2)$ at the respective ratios. The other recipient grid points are $(4, 0)$, $(0, 4)$, $(2, 0)$, $(2, 2)$ and are unpaired. Adding $1/8$ to each weight leaves every total below one, proving all three cases.

At ratio $4/3$, the two overloaded groups are

$$A^4 + BD^3/4096, \quad A^3C + D^4/65536.$$

Weighted AM–GM gives

$$BD^3/4096 + D^4/65536 \leq B^4/16384 + 13D^4/65536.$$

Keep A^4 in its group and combine the D^4 term with A^3C using (4.6)–(4.7), with $\epsilon = 13/65536$ and $X = D$. This adds $1/16384$ at B^4 and $13/65536$ at each of A^2D^2, A^2C^2 . Their distinct recipient groups have original weights $1/256, 9/64, 3/8$, so all additions fit.

This proves every tensor comparison. For positive A, B, C, D , it is strict: the group containing A^2B^2 has a positive maximum and final weight below one. Its original weight is $3/8$; the outer-pivot transfers leave it at $217/512$ or $97/256$, the first three inner-pivot transfers leave it at $1/2$, and the inner ratio $4/3$ leaves it unchanged. Together with the twelve three-corner profiles, this proves Proposition 4.1. The finite maximum in (4.2) preserves strictness.

5 Four-term arithmetic progressions

Let $F = \{0, 1, 2, 3\}$. For $k = (k_0, k_1, k_2, k_3) \geq 0$, write

$$T(k) = T_F(k) = \sum_{s=0}^{12} \max_{i_1+i_2+i_3+i_4=s} \prod_{r=1}^4 k_{i_r}, \quad i_r \in F.$$

For a tuple v , let $v_i^R = v_{3-i}$. Define the five squared profiles

$$\begin{aligned} A &= (1, 1/4, 0, 1/4), \\ B &= (1, 0, 1/4, 1/4), \\ C &= (1, 1/4, 9/64, 25/256), \\ D &= (1/4, 1, 0, 1/4), \\ E &= (0, 1, 1/4, 9/64), \end{aligned}$$

and the ten-element set

$$\mathcal{V} = \{A, B, C, D, E, A^R, B^R, C^R, D^R, E^R\}.$$

Proposition 5.1. *The weighted norm has the finite-profile formula*

$$\sup_{x \geq 0, x * x \leq 1} \sum_{i=0}^3 k_i x_i^2 = \max_{v \in \mathcal{V}} k \cdot v, \quad (5.1)$$

where the convolution inequality is coefficientwise at all seven indices, and

$$(k \cdot v)^4 \leq T(k) \quad (v \in \mathcal{V}). \quad (5.2)$$

Consequently,

$$x * x \leq 1 \implies \left(\sum_i k_i x_i^2 \right)^4 \leq T(k) \quad (k \geq 0). \quad (5.3)$$

For positive k_i , comparison (5.2), and hence (5.3) at the maximizing value, is strict. These statements hold on every four-term integer arithmetic progression after relabeling the coordinates.

5.1 Feasibility and a six-profile convex hull

Each member of $\mathcal{V}$ is the square of a feasible tuple. For the five unreversed members, the seven coefficients of $\sqrt{v} * \sqrt{v}$ are

Profile	Ordinary convolution coefficients
A	$(1, 1, 1/4, 1, 1/2, 0, 1/4)$
B	$(1, 0, 1, 1, 1/4, 1/2, 1/4)$
C	$(1, 1, 1, 1, 29/64, 15/64, 25/256)$
D	$(1/4, 1, 1, 1/2, 1, 0, 1/4)$
E	$(0, 0, 1, 1, 1, 3/8, 9/64)$

Reversal reverses this coefficient list. Thus the right-hand side of (5.1) is always attained by a feasible squared profile.

Let

$$\mathcal{S} = \{A, B, D, A^R, B^R, D^R\}.$$

Its convex hull has the exact elementary description

$$\text{conv}(\mathcal{S}) = \left\{ y \geq 0 : \sum_i y_i = \frac{3}{2}, \quad y_0, y_3 \geq \frac{1}{4}, \quad y_0 + y_3 \leq \frac{5}{4} \right\}. \quad (5.4)$$

To check the reverse inclusion, put

$$r_0 = \frac{4y_0 - 1}{3}, \quad r_3 = \frac{4y_3 - 1}{3}, \quad r_m = 1 - r_0 - r_3.$$

These are nonnegative and sum to one. Allocate the same fraction $p = y_1/(y_1 + y_2)$ to the first member of each pair (A, B) , (B^R, A^R) , and (D, D^R) , with pair masses r_0, r_3, r_m , respectively. The resulting convex combination is y . The denominator is positive because $y_1 + y_2 \geq 1/4$.

A useful consequence is the following sufficient domination test. If a tuple $b \geq 0$ satisfies

$$b_0, b_3 \leq \frac{1}{2}, \quad m = b_1 + b_2 \leq 1,$$

$$m + \max(b_0, 1/4) + \max(b_3, 1/4) \leq \frac{3}{2}, \quad (5.5)$$

then some $y \in \text{conv}(\mathcal{S})$ dominates b coordinatewise. Indeed choose the final middle sum to be $m' = \max(m, 1/4)$, and the endpoint sum to be $3/2 - m'$. The latter lies in $[1/2, 5/4]$ and is at least $\max(b_0, 1/4) + \max(b_3, 1/4)$: this is (5.5) when $m \geq 1/4$, and follows from $b_0, b_3 \leq 1/2$ otherwise. Increase coordinates to these sums and apply (5.4).

5.1.1 Self-contained proof of the sharp norm lemma

Let $N = \sum_i x_i^2$ and $M = \max_i x_i^2$. The constraints imply $x_i^2 \leq 1$, $2x_i x_j \leq 1$ for $i \neq j$, and $x_0 x_3 + x_1 x_2 \leq 1/2$. We repeatedly use

$$0 \leq r, s \leq J \implies r + s \leq J + rs/J, \quad (5.6)$$

which follows from $(J - r)(J - s) \geq 0$.

If $M \leq 1/2$, put $r = x_0 x_3$, $s = x_1 x_2$. Applying (5.6) to the squared coordinates in each pair yields

$$N \leq 1 + 2r^2 + 2s^2 \leq 1 + 2(r + s)^2 \leq 3/2.$$

Suppose $1/2 \leq M \leq 3/4$, attained at index i . Put $y = x_i x_{3-i}$, $J = 1/(4M)$, and let r be the product of the two coordinates outside this pair. The other three squared coordinates are at most J , while $r + y \leq 1/2$ and $r \leq J$. Hence

$$N \leq M + y^2/M + J + 4M \min(J, 1/2 - y)^2.$$

For $0 \leq y \leq 1/2 - J$, this expression increases with y . For $1/2 - J \leq y \leq 1/2$, it is a convex quadratic. Its maximum is therefore the larger of the two endpoint values

$$B(M) = M + \frac{1}{2M} + \frac{(2M - 1)^2}{16M^3}, \quad M + \frac{1}{2M},$$

namely $B(M)$. With $w = 2M - 1 \in [0, 1/2]$,

$$16M^3(3/2 - B(M)) = w(1 - w^2 - w^3) \geq 0.$$

Suppose $M \geq 3/4$ is at index one; index two follows by reversal. Put $J = 1/(4M)$. The constraint $M + 2x_0 x_2 \leq 1$ and (5.6) imply

$$x_0^2 + x_2^2 \leq J + M(1 - M)^2, \quad N \leq M + \frac{1}{2M} + M(1 - M)^2.$$

The difference from $3/2$ is

$$\frac{(1 - M)(2M^3 - 2M^2 + 2M - 1)}{2M} \geq 0.$$

The cubic increases, since its derivative is $6(M - 1/3)^2 + 4/3 > 0$, and is $7/32$ at $M = 3/4$.

Finally suppose an endpoint has squared value $u \in [1/2, 1]$. By reversal take $x_0^2 = u$, and put $a = 2x_0 x_1$, $b = 2x_0 x_2$, $c = 2x_0 x_3$. The first four constraints give

$$0 \leq a \leq 1, \quad 0 \leq b \leq 1 - a^2/4, \quad 0 \leq c \leq 1 - ab/2.$$

For fixed a , the upper bound $a^2 + b^2 + (1 - ab/2)^2$ on $a^2 + b^2 + c^2$ is convex in b . At $b = 0$, it is at most two. At $b = 1 - a^2/4$, it is

$$\Psi(a) = 2 - a + 3a^2/4 + a^3/4 - a^4/16 + a^6/64.$$

This is convex on $[0, 1]$, since $\Psi''(a) = 3/2 + 3a/2 - 3a^2/4 + 15a^4/32 \geq 3/4$, and its endpoint values are 2 and $125/64 < 2$. Therefore

$$N = u + \frac{a^2 + b^2 + c^2}{4u} \leq u + \frac{1}{2u} \leq \frac{3}{2}.$$

These cases cover every feasible tuple. The norm bound is sharp at $x = (1/2, 1, 0, 1/2) = \sqrt{D}$, as its feasibility table shows.

5.2 Reduction to the ten profiles

For a feasible x , write $b_i = x_i^2$. The convolution constraints give

$$b_i \leq 1, \quad b_i b_j \leq \frac{1}{4} \quad (i \neq j). \quad (5.7)$$

We also use the previously proved sharp norm lemma

$$\sum_i b_i \leq \frac{3}{2}. \quad (5.8)$$

5.2.1 All squared coordinates at most one half

Suppose $b_i \leq 1/2$ for all i . Each consecutive triple has sum at most $9/8$. For example, the constraint $b_1 + 2\sqrt{b_0 b_2} \leq 1$, together with $b_0, b_2 \leq 1/2$, gives

$$\begin{aligned} b_0 + b_2 &\leq \frac{1}{2} + 2b_0 b_2 \leq \frac{1}{2} + \frac{(1 - b_1)^2}{2}, \\ b_0 + b_1 + b_2 &\leq 1 + \frac{b_1^2}{2} \leq \frac{9}{8}. \end{aligned} \quad (5.9)$$

The other triple is the reversed argument.

Condition (5.5) now holds. If both endpoints exceed $1/4$, use (5.8); if both are below $1/4$, use $b_1 + b_2 \leq 1$. If just one is below $1/4$, the expression in (5.5) is a consecutive triple sum plus $1/4$, at most $11/8 < 3/2$. Thus b is dominated by the convex hull of $\mathcal{S}$.

5.2.2 A middle squared coordinate at least one half

By reversal, it suffices to write

$$b = (a, u, v, d), \quad 1/2 \leq u \leq 1.$$

Put $J = 1/(4u)$. By (5.7), $a, v, d \leq J \leq 1/2$. The constraint at index two gives $\sqrt{av} \leq (1 - u)/2$. Consequently

$$a + v \leq J + \frac{av}{J} \leq \frac{1}{4u} + u(1 - u)^2.$$

The following bound is useful throughout this case:

$$a + u + v \leq \frac{5}{4}. \quad (5.10)$$

Indeed

$$\frac{5}{4} - \left(u + \frac{1}{4u} + u(1-u)^2 \right) = \frac{(1-u)(4u^3 - 4u^2 + 4u - 1)}{4u} \geq 0.$$

The cubic is increasing, because $12u^2 - 8u + 4 > 0$, and its value at $u = 1/2$ is $1/2$.

First suppose $m = u + v \leq 1$. The domination test (5.5) holds. The cases when both endpoints are on the same side of $1/4$ follow from (5.8) or $m \leq 1$. If $a \geq 1/4$ and $d \leq 1/4$, use (5.10). If $a \leq 1/4$ and $d \geq 1/4$, the index-four constraint gives

$$u + v + d \leq 1 + (\sqrt{u} - \sqrt{d})^2 \leq \frac{5}{4},$$

because $d \leq 1/2 \leq u \leq 1$ and $d \geq 1/4$, so $0 \leq \sqrt{u} - \sqrt{d} \leq 1/2$. Thus (5.5) again proves domination by $\text{conv}(\mathcal{S})$.

Now suppose $m = u + v > 1$. Define

$$\theta = 4(m-1), \quad \beta = 4 - 3u - 4v, \quad \gamma = 1 - u.$$

All three numbers are nonnegative and their sum is one. For β , use $v \leq 1/(4u)$ to obtain

$$\beta \geq 4 - 3u - 1/u = \frac{(1-u)(3u-1)}{u} \geq 0.$$

The convex combination

$$y = \theta E + \beta D + \gamma D^R = \left(\frac{5}{4} - m, u, v, \frac{11-7m}{16} \right) \quad (5.11)$$

dominates b . The first coordinate does so by (5.10). For the last, the index-four constraint gives $d \leq (1-v)^2/(4u)$. We claim

$$\frac{(1-v)^2}{4u} \leq \frac{11-7u-7v}{16} \quad \left(1-u \leq v \leq \frac{1}{4u} \right). \quad (5.12)$$

After multiplying by $16u$, the required nonnegative expression is

$$G_u(v) = 11u - 7u^2 - 7uv - 4 + 8v - 4v^2.$$

It is concave in v , so its minimum on this interval is at an endpoint. The two endpoint values are

$$G_u(1-u) = 4u(1-u),$$

$$G_u(1/(4u)) = \frac{(1-u)(28u^3 - 16u^2 + 7u - 1)}{4u^2}.$$

The latter cubic is increasing on $[1/2, 1]$, since its derivative $84u^2 - 32u + 7$ is positive there, and its value at $1/2$ is two. This proves (5.12), then (5.11), and completes the middle-coordinate case.

5.2.3 An endpoint squared coordinate at least one half

By reversal, assume $u = x_0^2 \in [1/2, 1]$. Put

$$\alpha = 2x_0x_1, \quad \beta = 2x_0x_2, \quad \gamma = 2x_0x_3.$$

The first four convolution constraints give

$$0 \leq \alpha \leq 1, \quad 0 \leq \beta \leq 1 - \alpha^2/4, \quad 0 \leq \gamma \leq 1 - \alpha\beta/2.$$

For arbitrary nonnegative k ,

$$k \cdot b = k_0u + \frac{k_1\alpha^2 + k_2\beta^2 + k_3\gamma^2}{4u}.$$

For fixed α , the numerator is at most

$$k_1\alpha^2 + k_2\beta^2 + k_3(1 - \alpha\beta/2)^2,$$

which is convex in β . At $\beta = 0$, it is at most $k_1 + k_3$. At $\beta = 1 - \alpha^2/4$, it is the polynomial

$$\begin{aligned} \Phi(\alpha) &= k_2 + k_3 - k_3\alpha + (k_1 - k_2/2 + k_3/4)\alpha^2 + k_3\alpha^3/4 \\ &\quad + (k_2/16 - k_3/8)\alpha^4 + k_3\alpha^6/64. \end{aligned}$$

This polynomial has its maximum on $[0, 1]$ at an endpoint, even when it is not convex. To see this, observe

$$\Phi'(0) = -k_3 \leq 0,$$

$$\Phi'''(\alpha) = \frac{3}{2}k_2\alpha + \frac{3}{8}k_3(4 - 8\alpha + 5\alpha^3) \geq 0.$$

For the last inequality, the minimum of the cubic on $[0, 1]$ is at $\alpha = \sqrt{8/15}$, where its value is $4 - (16/3)\sqrt{8/15} \geq 0$, since $\sqrt{8/15} \leq 3/4$. Thus Φ' is convex and starts nonpositive; its nonpositive sublevel set is an interval containing zero. Accordingly Φ first decreases and then increases, with either portion possibly absent. Its maximum is at an endpoint.

The two endpoint values of Φ are $k_2 + k_3$ and $k_1 + 9k_2/16 + 25k_3/64$. Therefore the numerator is at most

$$H = \max\{k_1 + k_3, k_2 + k_3, k_1 + 9k_2/16 + 25k_3/64\}.$$

For each of these three linear forms, convexity in $u \in [1/2, 1]$ reduces $k_0u + H/(4u)$ to its two endpoint values. At $u = 1$, the resulting squared profiles are A, B, C . At $u = 1/2$, they are

$$W_A = (1/2, 1/2, 0, 1/2), \quad W_B = (1/2, 0, 1/2, 1/2),$$

$$W_C = (1/2, 1/2, 9/32, 25/128).$$

The first two belong to the sparse convex hull, since

$$W_A = (A + D + B^R)/3, \quad W_B = (B + D^R + A^R)/3.$$

The last is dominated coordinatewise by

$$\frac{1}{3}C + \frac{5}{12}D + \frac{1}{4}D^R = \left(\frac{1}{2}, \frac{1}{2}, \frac{19}{64}, \frac{51}{256}\right) \geq W_C.$$

Consequently $k \cdot b \leq \max_{v \in \mathcal{V}} k \cdot v$ in this case as well. Every feasible tuple falls into one of the three cases. Together with the explicit feasibility table, this proves (5.1).

5.3 Comparing the ten linear forms with the four-fold maximum

Reversal preserves T , so it suffices to handle A, B, C, D, E .

5.3.1 A coefficient lemma; profiles C and E

If $v \geq 0$ has the property that every coefficient of $(\sum_i v_i z^i)^4$ is at most one, then

$$(k \cdot v)^4 = \sum_s \sum_{i_1 + \dots + i_4 = s} \prod_r (v_{i_r} k_{i_r}) \leq \sum_s \max_{i_1 + \dots + i_4 = s} \prod_r k_{i_r} = T(k). \quad (5.13)$$

The nonnegative formal series

$$(1 - z)^{-1/4} = 1 + z/4 + 5z^2/32 + 15z^3/128 + \dots$$

coefficientwise dominates $1 + z/4 + 9z^2/64 + 25z^3/256$. Raising to the fourth power preserves this order, and $((1 - z)^{-1/4})^4 = (1 - z)^{-1}$ has all coefficients equal to one. This proves the required coefficient property for C .

For E , use $\sum_i E_i z^i = z(1 + z/4 + 9z^2/64)$. The same comparison applies before the harmless fourth-power shift by four. Thus (5.13) proves both profiles.

5.3.2 Profile B

Set $a = k_0$, $b = k_2$, $c = k_3$, and ignore k_1 ; doing so can only decrease T . The target linear form is $a + b/4 + c/4$, on indices 0, 2, 3. Grouping its fourth-power expansion by index sum, the only fibers with more than one monomial are

Index sum	Monomials and their expansion weights	Total weight
6	$ab^3/16 + 3a^2c^2/8$	7/16
8	$b^4/256 + 3abc^2/16$	49/256
9	$b^3c/64 + ac^3/16$	5/64

Every other fiber has one monomial with weight at most one. Each fiber is thus bounded by its largest kernel monomial. Summing proves the comparison for B .

5.3.3 The common three-point maximum for A and D

Now put $a = k_0$, $b = k_1$, $c = k_3$, and ignore k_2 . On the indices $\{0, 1, 3\}$, the four-fold maximum sum is exactly

$$\begin{aligned} \mathcal{T}(a, b, c) = & a^4 + a^3b + a^2b^2 + \max(a^3c, ab^3) + \max(a^2bc, b^4) + ab^2c \\ & + \max(a^2c^2, b^3c) + abc^2 + b^2c^2 + ac^3 + bc^3 + c^4. \end{aligned} \quad (5.14)$$

These terms correspond to sums 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, respectively; sum eleven is absent. In particular $T(k) \geq \mathcal{T}(a, b, c)$.

For A , let

$$P_A = a^4 + a^3b + a^3c + a^2b^2 + \frac{3}{4}a^2bc + \frac{1}{4}b^4 + ab^2c \\ + \frac{1}{2}a^2c^2 + \frac{1}{2}b^3c + abc^2 + b^2c^2 + ac^3 + bc^3 + c^4.$$

Selecting one term or a convex average in each maximum in (5.14) gives $\mathcal{T} \geq P_A$. An exact nonnegative remainder is

$$P_A - (a + b/4 + c/4)^4 = \frac{b^2(a-b)^2}{32} + \frac{19a^2b^2}{32} + \frac{a^2c^2}{8} \\ + \frac{13ab^2c}{16} + \frac{13abc^2}{16} + \frac{15ac^3}{16} + \frac{55b^4}{256} \\ + \frac{31b^3c}{64} + \frac{125b^2c^2}{128} + \frac{63bc^3}{64} + \frac{255c^4}{256}. \quad (5.15)$$

Thus (5.2) holds for A .

For D , choose the terms ab^3, b^4, b^3c in the three maxima of (5.14), obtaining the lower bound

$$P_D = a^4 + a^3b + a^2b^2 + ab^3 + b^4 + ab^2c + b^3c + abc^2 + b^2c^2 + ac^3 + bc^3 + c^4.$$

In comparison with $(a/4 + b + c/4)^4$, the only missing monomials are

$$\frac{a^3c}{64}, \quad \frac{3a^2bc}{16}, \quad \frac{3a^2c^2}{128}.$$

The arithmetic–geometric mean inequalities

$$a^3c \leq \frac{3a^4 + c^4}{4}, \quad a^2bc \leq \frac{a^4 + b^2c^2}{2}, \quad a^2c^2 \leq \frac{a^4 + c^4}{2}$$

absorb their total using at most

$$\frac{30}{256}a^4 + \frac{4}{256}c^4 + \frac{3}{32}b^2c^2.$$

The available coefficient surpluses of these three monomials in $P_D - (a/4 + b + c/4)^4$ are respectively $255/256, 255/256, 5/8$, all larger than the required amounts. All other coefficient surpluses are nonnegative. This proves the comparison for D , and completes (5.2).

For positive k_i , these comparisons are strict. The profile C has a strict coefficient deficit at index sum two. Profiles A, D omit the positive sum-eleven contribution involving k_2 ; B omits the positive sum-one contribution involving k_1 ; and E omits the positive sum-zero contribution. Reversal handles the other five profiles.

6 One three-term progression

Let

$$F = \{0, a, 2a, b\}, \quad a > 0,$$

where the four integers are distinct and the only nontrivial equality of pair sums is $0 + 2a = a + a$. Equivalently, $q = b/a$ is not any of

$$-2, -1, 0, 1/2, 1, 3/2, 2, 3, 4. \quad (6.1)$$

Coordinates are $(L, M, R, O) = (0, a, 2a, b)$.

Let $\mathcal{V}$ consist of the following eight squared profiles:

$$\begin{aligned} V_L &= (1, 1/4, 9/64, 1/4), & V_R &= (9/64, 1/4, 1, 1/4), \\ D_L &= (1/4, 1, 0, 1/4), & D_R &= (0, 1, 1/4, 1/4), \\ E_L &= (1, 0, 1/4, 1/4), & E_R &= (1/4, 0, 1, 1/4), \\ O_1 &= (1/4, 1/4, 1/4, 1), & O_3 &= (1/3, 1/3, 1/3, 3/4). \end{aligned} \tag{6.2}$$

Proposition 6.1. *For arbitrary $k \geq 0$, the weighted norm is*

$$\sup_{x \cdot x \leq 1} \sum_{p \in F} k(p) x(p)^2 = \max_{v \in \mathcal{V}} k \cdot v. \tag{6.3}$$

In particular the sharp unweighted norm is $7/4$, attained by O_1, O_3 . Each listed profile satisfies

$$(k \cdot v)^4 \leq T_F(k) \quad \text{for all eight profiles,} \tag{6.4}$$

strictly when all four kernel values are positive.

6.1 Exact weighted norm reduction

Write $w_p = x(p)^2$. Ordinary feasibility on this support is exactly the individual bounds $w_p \leq 1$, pair bounds $w_p w_q \leq 1/4$, and

$$w_M + 2\sqrt{w_L w_R} \leq 1. \tag{6.5}$$

The pair bounds within the collided fiber are redundant but valid. Every profile in (6.2) is feasible, by direct substitution in (6.5) and the pair bounds.

We use the following elementary maximization repeatedly. For $0 \leq l, r \leq J$, $lr \leq c \leq J^2$, the maximum of any nonnegative linear objective occurs at $(J, c/J)$ or $(c/J, J)$. On the part of the boundary with $r = c/l$, the objective is convex in l ; elsewhere one coordinate can be increased. When $c \geq J^2$, use (J, J) .

First suppose every $w_p \leq 1/2$. Set $w_O = 1/2$. At fixed $m = w_M$, the endpoint rule gives an endpoint $1/2$, the other $(1-m)^2/2$. The objective is convex in $m \in [0, 1/2]$, so this region reduces to

$$H_0 = (1/2, 0, 1/2, 1/2), \quad H_L = (1/2, 1/2, 1/8, 1/2), \quad H_R, \tag{6.6}$$

where H_R exchanges the endpoints of H_L .

Otherwise let $u \in [1/2, 1]$ be a largest coordinate and put $J = 1/(4u)$. All other coordinates are at most J , so all their mutual pair bounds are automatic.

If the largest coordinate is L , set $O = J$. For fixed $M = m$, the best right endpoint is $J(1-m)^2$. Convexity in $m \in [0, J]$ reduces to

$$(u, 0, J, J), \quad (u, J, J(1-J)^2, J). \tag{6.7}$$

The first objective is convex in u , giving H_0, E_L . For the second, set

$$C = (1/u - 1)/3, \quad B = \frac{64}{55}(3/2 - u - 1/(2u)), \quad A = 1 - 2C - B.$$

These coefficients are nonnegative: $B = 32(2u-1)(1-u)/(55u)$, and $A = (192u^2 - 13u - 14)/(165u) > 0$ on this interval. The convex combination

$$AV_L + BV_R + CD_L + CO_1$$

matches the L, M, O coordinates of the second profile in (6.7). Its right coordinate is $317/192 - u - 49/(96u)$. The excess over $J(1 - J)^2$, multiplied by $192u^3$, is

$$(1 - u)(192u^3 - 125u^2 + 21u - 3) \geq 0.$$

The cubic is $1/4$ at $u = 1/2$ and increases: its derivative is $576u^2 - 250u + 21$, positive there and increasing thereafter. The case of largest coordinate R is reflected. This also dominates H_L, H_R by setting $u = 1/2$.

If $M = u$ is largest, set $O = J$. The endpoint rule gives $(L, R) = (J, u(1 - u)^2)$, or its reflection. Put the larger endpoint kernel weight on J , say $k_L \geq k_R$. The objective is

$$k_M u + k_R [J + u(1 - u)^2] + (k_L - k_R + k_O)J.$$

It is convex in u . Besides $J'' > 0$, the needed derivative is $[J + u(1 - u)^2]'' = 1/(2u^3) + 6u - 4 > 0$: split at $u = 2/3$, and use $1/(2u^3) \geq 27/16$, $6u - 4 \geq -1$ on the lower interval. The endpoints give H_L, D_L , or their reflections.

If $O = u$ is largest and $u \geq 3/4$, the best remaining coordinates are all J ; the objective is convex in u , giving O_3, O_1 . For $1/2 \leq u \leq 3/4$, optimize at fixed midpoint and use convexity in that midpoint. The only profiles needed are

$$(J, 1 - 2J, J, u), \quad (J, J, u - 1/2 + J/4, u), \quad (6.8)$$

and the reflected second profile. The first is dominated by a convex combination of H_0, O_3 : use weight $3(1 - 2J)$ on O_3 . This matches the first three coordinates and gives outsider coordinate $5/4 - 3J/2 \geq u$, since $(u - 1/2)(u - 3/4) \leq 0$. The second objective in (6.8) is a nonnegative multiple of $1/u$ plus a linear function, hence convex, giving H_L, O_3 .

Finally $H_0 \leq (E_L + E_R + O_1)/3$ coordinatewise. This completes the upper bound in (6.3); feasibility proves equality. The profile sums, in the order listed, are

$$105/64, 105/64, 3/2, 3/2, 3/2, 3/2, 7/4, 7/4.$$

6.2 The four triangle profiles

Each support triangle of D_L, D_R, E_L, E_R is 2-sum-Sidon: an additional midpoint relation would violate the isolated-AP3 assumption. For a nonnegative kernel on any 2-sum-Sidon triangle, let a be its largest value and R the sum of the other two. Fixing two copies of the largest point in the four-fold maximum gives

$$T(k) \geq a^4 + a^3 R + a^2 h_2(\text{other two values}) \geq a^4 + a^3 R + \frac{2}{3} a^2 R^2.$$

Here h_2 is the sum of the squares and pair product. Since $R \leq 3a$,

$$(a + R/4)^4 \leq a^4 + a^3 R + \frac{153}{256} a^2 R^2.$$

The difference is at least $53a^2 R^2/768$, positive when two values are positive. The largest pivot maximizes the unit-and-two-quarter linear form. Restricting k to the triangle and using monotonicity proves strict (6.4) for these four profiles when the full kernel is positive.

6.3 Common quartic framework and collision inventory

Write kernel values as $(A, B, C, D) = (k_L, k_M, k_R, k_O)$. In the fourth power of a weighted linear form, use triangular grid coordinates

$$i = n_B + 2n_C, \quad j = n_D, \quad 0 \leq j \leq 4, \quad 0 \leq i \leq 2(4 - j).$$

All monomials in this grid point have actual sum $ia + jb$. If the total nonnegative coefficient weights of a projected fiber are at most one, its weighted polynomial is bounded by that fiber's kernel-product maximum. Summing gives the required quartic comparison. The argument also allows stated AM–GM redistributions to other fibers before applying this test.

For O_1, O_3 , reflection about the AP midpoint permits $q = b/a > 1$. Write $q = r/s$ in lowest terms. Additional grid collisions differ by $(r, -s)$, so require $r \leq 8, s \leq 4$. Removing the forbidden ratios in (6.1) leaves precisely

s	r
1	5, 6, 7, 8
2	5, 7
3	4, 5, 7, 8
4	5, 7.

(6.9)

Each projected fiber contains at most two grid points. Ratios outside this list have no additional fourth-sum collision.

For the endpoint profile V_L , also consider its reflection V_R by allowing $q < 1$ in the same template. The additional positive ratios are $1/3, 2/3, 1/4, 3/4$. For $q = -r/s < 0$, a collision increases both grid coordinates by (r, s) , so $r + 2s \leq 8$. The allowed negative ratios are

$$-3, -4, -5, -6, -1/2, -3/2, -1/3, -2/3. \quad (6.10)$$

Again no fiber has more than two grid points. These lists exhaust the reflected cases as well: reflection replaces q by $2 - q$.

We repeatedly use

$$U + \epsilon V \leq \max(U, V) + \epsilon \min(U, V) \quad (0 \leq \epsilon \leq 1) \quad (6.11)$$

and

$$\min(X^3Y, Z^4) \leq X^2Z^2 + X^2Y^2. \quad (6.12)$$

For (6.12), divide by X^4 when $X > 0$. The assertion $\min(y, z^4) \leq z^2 + y^2$ follows by splitting into $z \leq 1, z \geq 1, y \leq 1$, and $y \geq 1$. The zero case is immediate.

6.4 The profile O3

Here $E = (A + B + C)/3 + 3D/4$. Let

$$P_2 = (1, 2, 3, 2, 1), \quad P_3 = (1, 3, 6, 7, 6, 3, 1),$$

$$P_4 = (1, 4, 10, 16, 19, 16, 10, 4, 1)$$

be the coefficients of $(1 + z + z^2)^2, (1 + z + z^2)^3, (1 + z + z^2)^4$. The grid coefficient rows of E^4 are

$$j = 0 : P_4/81; \quad j = 1 : P_3/9; \quad j = 2 : 3P_2/8; \\ j = 3 : (9/16, 9/16, 9/16); \quad j = 4 : (81/256).$$

At grid (2, 2), move weight $1/4$ of ACD^2 using $ACD^2 \leq (A^2D^2 + C^2D^2)/2$. The new $j = 2$ row is

$$(1/2, 3/4, 7/8, 3/4, 1/2).$$

All other rows are unchanged. Uncollided fibers have weight at most $7/8$. For the four denominator rows of (6.9), upper bounds on collided-fiber weights are respectively

$$31/36, \quad 599/648, \quad 755/768, \quad 81/256 + 16/81,$$

all below one. These follow by adding the displayed rows with displacement $(r, -s)$. For example the closest case is $r/s = 4/3$, where grid (0, 4) meets (4, 1), giving $81/256 + 2/3 = 755/768$. The coefficient test proves (6.4), with strictness for every nonzero kernel.

6.5 The profile O1

Now $E = D + (A + B + C)/4$. Its coefficient rows are

$$j = 0 : P_4/256; \quad j = 1 : P_3/16; \quad j = 2 : 3P_2/8; \quad j = 3 : (1, 1, 1); \quad j = 4 : (1).$$

Move weight $1/2$ of ACD^2 by the same AM–GM inequality. The $j = 2$ row becomes

$$(5/8, 3/4, 5/8, 3/4, 5/8). \tag{6.13}$$

For ratios $5, 6, 7, 8, 7/2$, every projected fiber now has weight at most one. At the other ratios, retain the full $j = 3$ or $j = 4$ polynomial in each overloaded fiber and remove its other grid-point polynomial. The following ledgers absorb precisely those removed terms. All unmentioned fibers pass the coefficient test with (6.13).

At $q = 5/2$, the removed terms are

$$3DBC^2/16 + DC^3/16.$$

They are bounded by $D^2C^2/8 + 3B^2C^2/32 + C^4/32$. The original weights of these three projected fibers were $5/8, 101/128, 193/256$. After allocation they are $3/4, 113/128, 201/256$, below one.

At $q = 5/3$, the removed AP-only rows are at $i = 5, 6, 7$, with total coefficient weight $30/256$; the removed $j = 1, i = 5$ term is $3DBC^2/16$. Every removed unweighted monomial is at most

$$S = A^4 + B^4 + C^4 + D^2C^2,$$

by weighted AM–GM (for DBC^2 , use $DBC^2 \leq D^2C^2/2 + B^4/4 + C^4/4$). The total removed weight is $39/128$. The four S monomials occupy distinct fibers, of original weights $1/256, 19/256, 1/256, 5/8$. Adding $39/128$ to each remains below one; the largest becomes $119/128$.

At $q = 4/3$, the removed AP-only rows are $i = 4, 5, 6$, with total weight $45/256$. Bound them by $45(A^4 + B^4 + C^4)/256$. The removed $j = 1, i = 4$ polynomial is $3DAC^2/16 + 3DB^2C/16$, bounded by

$$3D^2C^2/16 + 3A^4/64 + 3C^4/64 + 3B^4/32.$$

Thus the allocations are $57/256$ each to A^4, C^4 , $3/16$ to D^2C^2 , and $69/256$ to B^4 . The last shares the fiber of D^3A . Apply (6.11)-(6.12) to $D^3A + (69/256)B^4$; the remainder is at most $69(D^2B^2 + D^2A^2)/256$. The final affected weights are $58/256$ at each of A^4, C^4 , $13/16$ at D^2C^2 , and $229/256$ at each of D^2B^2, D^2A^2 . All are below one.

At $q = 7/3$, the removed terms are $BC^3/64 + C^4/256$, at the fibers of D^3A, D^3B , respectively. They are bounded by $B^4/256 + C^4/64$. Use (6.11)-(6.12) for $D^3B + C^4/64$, allocating its remainder to $(D^2C^2 + D^2B^2)/64$. The B^4 fiber had weight $19/256$; the two quadratic fibers had weight $5/8$. These additions fit. At $q = 8/3$, only $C^4/256$ is removed, at the D^3A fiber; (6.11)-(6.12) allocate its remainder to $(D^2C^2 + D^2A^2)/256$.

At $q = 5/4$, the removed AP-only row is $i = 5$, of total weight $1/16$, at the D^4 fiber. Each monomial is at most $A^4 + B^4 + C^4$. These three distinct unoverloaded fibers have weights $1/256, 19/256, 1/256$; adding $1/16$ fits. At $q = 7/4$, the sole removed term is $BC^3/64$, absorbed into $(B^4 + C^4)/64$ in the same way. This exhausts (6.9). Every ledger leaves positive slack in a quadratic fiber, so (6.4) is strict for positive kernels.

6.6 Endpoint profiles: coefficients and positive ratios

It suffices to study V_L at all ratios in (6.9)-(6.10) and the four positive ratios below one. Reflection then gives V_R . Write

$$E = A + B/4 + 9C/64 + D/4.$$

Its triangular coefficient rows, in increasing i , are

j	row
0	(1, 1, 15/16, 31/64, 467/2048, 279/4096, 1215/65536, 729/262144, 6561/16777216)
1	(1, 3/4, 39/64, 29/128, 351/4096, 243/16384, 729/262144)
2	(3/8, 3/16, 33/256, 27/1024, 243/32768)
3	(1/16, 1/64, 9/1024)
4	(1/256).

(6.14)

These are the coefficients of $(1 + z/4 + 9z^2/64 + w/4)^4$. Without extra collisions they all pass the coefficient test. For $q > 1$ in (6.9), the test also passes directly unless q is an integer 5, 6, 7, 8. At those integers the only overloaded fiber is A^3D , paired with AP grid point $(q, 0)$. Its additional polynomial is as follows:

q	additional polynomial
5	$243ABC^2/4096 + 9B^3C/1024$
6	$729AC^3/65536 + 243B^2C^2/32768$
7	$729BC^3/262144$
8	$6561C^4/16777216$.

(6.15)

For $q = 5$, use $ABC^2 \leq (A^2B^2 + C^4)/2$ and $B^3C \leq (3B^4 + C^4)/4$. The extra weight at A^2B^2 is $243/8192 < 1/16$, fitting its original weight $15/16$. The extra weights at B^4, C^4 are each below $1/16$; their original fiber weights are $467/2048$ and $29/128 + 6561/16777216$, respectively.

For $q = 6$, use $AC^3 \leq (A^2C^2 + C^4)/2$ and $B^2C^2 \leq (B^4 + C^4)/2$. The first and third terms named in these bounds, A^2C^2, B^4 , share AP fiber four. Its extra total weight, and the extra weight at C^4 , are both $1215/131072 < 1/64$. Their original weights are $467/2048$ and $39/64 + 6561/16777216$.

For $q = 7$, use $BC^3 \leq (B^4 + 3C^4)/4$; both extra allocations are below $1/256$, fitting original weights $467/2048$ and $3/4 + 6561/16777216$. For $q = 8$, apply (6.11)-(6.12) directly

to the overloaded fiber $A^3D + \epsilon C^4$, where $\epsilon = 6561/16777216 < 1/256$, allocating the remainder to $\epsilon(A^2C^2 + A^2D^2)$. Those fibers have original weights $467/2048$ and $3/8$.

For the positive ratios below one, $q = 2/3, 3/4$ pass directly using (6.14). At $q = 1/3$, the only overload is $A^3B + AD^3/16$. Allocate the second term to $(A^2D^2 + D^4)/32$, in fibers originally of weights $3/8$ and $3/4 + 1/256$. At $q = 1/4$, the sole overload is $A^3B + D^4/256$. Equations (6.11)-(6.12) allocate its remainder to $(A^2D^2 + A^2B^2)/256$, fitting original weights $3/8, 15/16$.

6.7 Endpoint profiles: negative ratios

The following table lists exactly the additional, high-grid-point polynomials removed from overloaded fibers. Retain their low-grid-point polynomials in full. All other fibers already pass the coefficient test with (6.14).

q	Removed polynomial
-3	$B^3D/64 + 27ABCD/128 + 27B^2CD/1024 + 243AC^2D/4096 + 27BCD^2/1024$
-4	$27B^2CD/1024 + 243AC^2D/4096 + 243BC^2D/16384 + 243C^2D^2/32768$
-5	$243BC^2D/16384 + 729C^3D/262144$
-6	$729C^3D/262144$
-1/2	$3ABD^2/16 + 3B^2D^2/128 + 27ACD^2/256 + BD^3/64$
-3/2	$27BCD^2/1024 + 243C^2D^2/32768$
-1/3	$BD^3/64 + 9CD^3/1024$
-2/3	$9CD^3/1024$

For $q = -3, -4$, every removed monomial is at most

$$S = A^2C^2 + B^4 + C^4 + D^4 + B^2D^2 + A^2D^2.$$

Use weighted AM–GM for monomials without A , and $ABCD \leq (A^2C^2 + B^2D^2)/2$, $AC^2D \leq (A^2D^2 + C^4)/2$. The removed total weights are respectively $1387/4096 < 11/32$ and $3537/32768 < 1/8$. Allocate $11/32$, or $1/8$, to each monomial of S . The first two share a fiber, originally of weight $467/2048$; even the larger addition gives $467/2048 + 2(11/32) = 1875/2048 < 1$. The other four occupy distinct fibers, originally of weights $6561/16777216, 1/256, 33/256, 3/8$, so all allocations fit.

For $q = -1/2$, use

$$S = A^2D^2 + B^4 + C^4 + D^4.$$

Every removed monomial is at most S : for example $ABD^2 \leq A^2D^2/2 + B^4/4 + D^4/4$, and the same bound with C replaces B for ACD^2 . The removed total weight is $85/256 < 3/8$. The four S monomials occupy distinct fibers whose original weights are

$$3/8, 467/2048, 6561/16777216, 1/256.$$

Adding $3/8$ to each fits.

For each remaining row of the table, every monomial is at most $B^4 + C^4 + D^4$ by weighted AM–GM. The removed total weights have the following convenient upper bounds:

q	-5	-6	-3/2	-1/3	-2/3
bound	$1/32$	$1/256$	$1/16$	$1/32$	$1/64$

These three pure fourth powers occupy distinct unoverloaded fibers, of original weights $467/2048, 6561/16777216, 1/256$, so the stated additions fit. This finishes (6.10) and hence both endpoint profiles.

The endpoint inequalities are strict for positive kernels. Except at $q = 2/3$, the AP-sum-two fiber still has total coefficient weight below one after the allocations. At $q = 2/3$, use the AP-sum-four fiber, whose weight is $467/2048 + 9/1024 < 1$. In each case its maximum is positive. Together with Sections 6.2, 6.4, and 6.5, this proves strict (6.4).

7 Two intersecting three-term progressions

The support $F = \{0, 1, 2, 4\}$ contains the progressions $\{0, 1, 2\}$ and $\{0, 2, 4\}$. Both relations enter the weighted estimate. For a nonnegative x supported on F , put $w_i = x(i)^2$. The condition $x * x \leq 1$ is equivalent to

$$\begin{aligned} 0 \leq w_i \leq 1, \quad w_i w_j \leq \frac{1}{4} \quad (i \neq j), \\ w_1 + 2\sqrt{w_0 w_2} \leq 1, \quad w_2 + 2\sqrt{w_0 w_4} \leq 1. \end{aligned} \quad (7.1)$$

All tuples in this section use the coordinate order $(0, 1, 2, 4)$. We shall bound nonnegative linear functionals of w by their values on the profiles

$$\begin{aligned} H_0 &= (\tfrac{1}{2}, 0, \tfrac{1}{2}, \tfrac{1}{2}), & H_L &= (\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{8}, \tfrac{1}{2}), \\ H_R &= (\tfrac{1}{8}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), & V_L &= (1, \tfrac{1}{4}, \tfrac{9}{64}, \tfrac{1}{4}), \\ V_R &= (\tfrac{9}{64}, \tfrac{1}{4}, 1, \tfrac{1}{4}), & R &= (\tfrac{9}{64}, \tfrac{1}{4}, \tfrac{1}{4}, 1), \end{aligned} \quad (7.2)$$

and on feasible profiles with a zero coordinate. The tuples H_0, H_L, V_L, V_R are used only as upper bounds; they need not satisfy (7.1).

7.1 Profiles with a zero coordinate

Lemma 7.1. *Suppose that w satisfies (7.1) and has a zero coordinate. For every nonnegative kernel k on F , put $k_w = k \mathbf{1}_{\text{supp}(w)}$. Then*

$$(k \cdot w)^4 \leq T_F(k_w) \leq T_F(k). \quad (7.3)$$

The last comparison is strict when k is positive on F . Moreover, $\sum_i w_i \leq 3/2$.

Proof. A three-term progression embeds in a four-term progression. Extending k and x by zero at the additional point preserves ordinary convolution feasibility and the two quantities in the weighted comparison of Section 5. This proves the first inequality when the remaining support is a progression.

For a triangle with no three-term progression, its pair sums are distinct apart from interchange. Write a for its largest kernel value and r for the sum of the other two. Its weighted squared norm is

$$a + \frac{r}{4}. \quad (7.4)$$

Indeed, if all squared coordinates are at most $1/2$, the half-triple dominates them and is the average of the three tuples having one coordinate equal to 1 and the other two equal to $1/4$. Otherwise, a largest squared coordinate $u \in [1/2, 1]$ bounds the other two by $1/(4u)$. The resulting linear functional is convex in u , so its maximum occurs at an endpoint. Choosing the largest kernel coordinate for the unit coordinate proves (7.4).

Fixing two copies of the point carrying a in the fourfold maximum gives six distinct sums and therefore

$$T_F(k_w) \geq a^4 + a^3 r + \frac{2}{3} a^2 r^2 \geq (a + r/4)^4. \quad (7.5)$$

For the final inequality, expand the fourth power. Since $r \leq 2a \leq 3a$, the quadratic and higher terms are at most $(153/256)a^2r^2 < (2/3)a^2r^2$ when $r > 0$. The zero cases follow directly. Supports of size at most two are covered by zero extension.

For a positive full kernel, a missing point 0, 4, 1, or 2 excludes the sum 0, 16, 1, or 14, respectively, from the smaller fourfold sumset. That sum has a positive contribution for the full kernel. This proves strictness. Finally, the same pair-bound argument with all weights equal to one gives total at most 3/2 for any feasible three-point profile; an additional progression constraint can only decrease this maximum. $\square$

7.2 Reduction of the feasible profiles

An endpoint estimate will be used for maxima at the first and third coordinates.

Lemma 7.2. *For $u \in [1/2, 1]$, set*

$$J = \frac{1}{4u}, \quad h = J(1 - J)^2. \quad (7.6)$$

The curve (u, J, h, J) is dominated coordinatewise by the convex hull of H_L, V_L, V_R . The curve (h, J, u, J) is dominated coordinatewise by the convex hull of H_R, V_R, V_L .

Proof. Set

$$\begin{aligned} t = 2u - 1, \quad s = \frac{(2u - 1)(1 - u)}{4u}, \quad \beta = \frac{s}{2}, \\ \alpha = t + \frac{23}{32}\beta, \quad \gamma = 1 - \alpha - \beta. \end{aligned} \quad (7.7)$$

These weights are nonnegative and sum to one, since

$$\gamma = (1 - u) \left(2 - \frac{55(2u - 1)}{256u} \right) \geq 0.$$

In $\gamma H_L + \alpha V_L + \beta V_R$, the first coordinate equals u , the second and fourth equal $J + 201s/256$, and the third equals

$$\frac{7 + 2u}{64} + \frac{1815s}{4096}.$$

On the other hand,

$$h - \frac{7 + 2u}{64} = s \frac{u^2 + 5u - 1}{16u^2} \leq \frac{7s}{16}, \quad (7.8)$$

because $6u^2 - 5u + 1 = (3u - 1)(2u - 1) \geq 0$. The inequality $1815 > 1792$ proves domination of the third coordinate. Interchanging the first and third coordinates proves the second assertion. $\square$

Lemma 7.3. *For every nonnegative k , its linear functional on the profiles satisfying (7.1) is bounded above by its largest value on the six profiles in (7.2) and on the feasible profiles having a zero coordinate.*

Proof. First suppose all coordinates are at most 1/2. Drop the second progression constraint for an upper bound and increase w_4 to 1/2. For fixed $m = w_1 \in [0, 1/2]$, the first constraint gives $w_0 w_2 \leq (1 - m)^2/4$. Maximizing a nonnegative linear functional under this product bound and the upper bounds $w_0, w_2 \leq 1/2$ reduces to

$$(w_0, w_2) = \left(\frac{1}{2}, \frac{(1 - m)^2}{2} \right) \quad \text{or its interchange.}$$

To see this, on the product boundary the objective has the convex form $a\ell + cK/\ell$; it is largest at an endpoint of the allowed interval. The remaining objective is convex in m , so its endpoints give H_0, H_L, H_R .

Otherwise choose a largest coordinate $u \geq 1/2$. Each other coordinate is at most $J = 1/(4u)$. We consider its four possible locations.

Largest coordinate at index 1. Drop the second progression constraint and set $w_4 = J$. The same product-bound argument gives

$$(w_0, w_2) = (J, f) \quad \text{or its interchange,} \quad f = u(1-u)^2. \quad (7.9)$$

Assign the larger of $k(0), k(2)$ to J . The objective becomes

$$k(1)u + k_{\text{small}}(J + f) + (k_{\text{large}} - k_{\text{small}} + k(4))J.$$

It is convex in u , since $J'' > 0$ and

$$(J + f)'' = \frac{1}{2u^3} + 6u - 4 > 0.$$

On $[1/2, 2/3]$, the first summand is at least $27/16$ and the second at least -1 ; above $2/3$ both are positive. The endpoint profiles are H_L, H_R and the feasible tuples

$$(1/4, 1, 0, 1/4), \quad (0, 1, 1/4, 1/4).$$

Largest coordinate at index 0. Retain both progression constraints. For fixed $c = w_2 \in [0, J]$, increase the other two coordinates to

$$w_1 = \min\{J, 1 - \sqrt{c/J}\}, \quad w_4 = J(1 - c)^2. \quad (7.10)$$

The first formula changes at $c = h$. The objective is convex on each of $[0, h]$ and $[h, J]$: its nonaffine terms are a nonnegative quadratic and, on the second interval, a nonnegative multiple of $-\sqrt{c}$. The resulting endpoints are

$$(u, J, 0, J), \quad (u, J, h, J(1 - h)^2), \quad (u, 0, J, h). \quad (7.11)$$

All three satisfy (7.1). The first and third have a zero coordinate. The middle is dominated by (u, J, h, J) , to which Lemma 7.2 applies.

Largest coordinate at index 2. Use $f = u(1 - u)^2$ again. We have $f \leq h$: after multiplication by $4u$, the equivalent inequality $1 - J - 2u(1 - u) \geq 0$ becomes

$$(2u - 1)(4u^2 - 2u + 1) \geq 0.$$

For fixed $a = w_0 > 0$, the other admissible maxima are

$$w_1 = \min\{J, 1 - 2\sqrt{au}\}, \quad w_4 = \min\left\{J, \frac{(1 - u)^2}{4a}\right\}. \quad (7.12)$$

The objective is affine or convex on the intervals with endpoints $0, f, h, J$, since the nonaffine terms are nonnegative multiples of $1/a$ and $-\sqrt{a}$. On $[0, f]$ it increases with a . It is enough to use the feasible profiles

$$(f, J, u, J), \quad \left(h, J, u, \frac{(1 - u)^2}{4h}\right), \quad (J, 0, u, f). \quad (7.13)$$

When $u = 1$, the condition $w_0 w_4 = 0$ gives these same limiting endpoints directly on its two branches. The last profile has a zero coordinate. Since $h \geq f$, the middle one is dominated by (h, J, u, J) and hence by Lemma 7.2. The first is dominated by the chord from H_R to the feasible tuple $(0, 1/4, 1, 1/4)$, with weight $2u - 1$ on the latter. Its first coordinate is $(1 - u)/4 \geq u(1 - u)^2$; its second and fourth dominate J by convexity of $1/u$.

Largest coordinate at index 4. The second progression constraint gives $w_0 \leq J(1-w_2)^2$. Drop the first constraint for an upper bound. The convex quadratic lies below its endpoint chord, so the profile is dominated by a convex combination of

$$B(u) = (J, J, 0, u), \quad W(u) = (J(1-J)^2, J, J, u). \quad (7.14)$$

The weight on $W(u)$ is w_2/J . Both endpoints also satisfy the first progression constraint; for $W(u)$ its left side is $3J - 2J^2 \leq 1$ for $J \in [1/4, 1/2]$.

Convexity of $1/u$ bounds $B(u)$ by the chord joining the feasible tuples

$$Q = (\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}), \quad S = (\frac{1}{4}, \frac{1}{4}, 0, 1).$$

For $W(u)$, put $Z = (1, 0, 1/4, 9/64)$ and define

$$\lambda = \frac{(1-u)(64u+55)}{87u}, \quad \mu = \frac{(2u-1)(64u+23)}{87u}, \quad \nu = \frac{32(2u-1)(1-u)}{87u}. \quad (7.15)$$

The weights are nonnegative and sum to one. The tuple $\lambda H_R + \mu R + \nu Z$ has coordinates $w_1 = J$, $w_4 = u$, and $w_2 = J + \nu/4$. Its first coordinate exceeds $J(1-J)^2$ by

$$\frac{(1-u)(2u-1)(576u^2 - 145u + 29)}{1856u^3} \geq 0. \quad (7.16)$$

The quadratic is positive because its leading coefficient is positive and its discriminant is $21025 - 66816 < 0$. The profiles Z, Q, S are feasible and have a zero coordinate. This finishes all four cases, including ties. $\square$

7.3 Quartic comparisons

Write $k = (A, B, C, D)$ on $(0, 1, 2, 4)$. In the expansion of $(k \cdot v)^4$, group monomials according to the sum of their four support indices. Each group is at most its coefficient sum times the maximum defining $T_F(k)$ at that index. Thus coefficient sums at most one prove the desired comparison. The following computations use weighted AM–GM to move a term to other fourfold products when a coefficient sum exceeds one.

Lemma 7.4. For $v = H_0, H_L, H_R, R$ and $k \geq 0$,

$$(k \cdot v)^4 \leq T_F(k). \quad (7.17)$$

Each inequality is strict when k is positive on F .

Proof. For H_0 , the target is $(A + C + D)^4/16$ on the progression $\{0, 2, 4\}$. After dividing indices by two, its coefficient sums are

$$\frac{1}{16}(1, 4, 10, 16, 19, 16, 10, 4, 1).$$

Use $A^2D^2 \leq (A^3D + AD^3)/2$ on the entire term $(6/16)A^2D^2$. The coefficient sums at indices 2, 4, 6 become $13/16$ each; all others were at most one. This compares with the three-point maximum. The positive full kernel also has a contribution at the missing odd sum 1, giving strictness.

For H_L , its index polynomial is $(1 + z + z^2/4 + z^4)/2$. If c_n is the coefficient of its fourth power, the complete coefficient sums are

n	0	1	2	3	4	5	6	7	8
$4096c_n$	256	1024	1792	1792	2144	3520	3952	2576	2497

n	9	10	11	12	13	14	15	16
$4096c_n$	3264	2320	768	1120	1024	256	0	256

(7.18)

These follow by squaring the row $(1, 2, 3/2, 1/2, 33/16, 2, 1/2, 0, 1)$ and dividing by 16. Every coefficient is at most $247/256 < 1$.

For H_R , the linear form is $A/8 + (B + C + D)/2$. Its coefficient sums c_n are

$$\begin{array}{c}
 \begin{array}{cccccccccc}
 n & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\
 \hline
 256c_n & 1/16 & 1 & 7 & 28 & 71 & 124 & 172 & 224 & 262 \\
 \hline
 \end{array} \\
 \begin{array}{cccccccccc}
 n & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 \\
 \hline
 256c_n & 240 & 208 & 192 & 112 & 64 & 64 & 0 & 16 \\
 \hline
 \end{array}
 \end{array} \tag{7.19}$$

Only sum 8 exceeds one. Replace its term $(3/128)A^2D^2$ by $(3/256)A^3D + (3/256)AD^3$. The coefficient sums at indices 4, 8, 12 become $74/256, 1, 115/256$, respectively. All others pass unchanged. The coefficient $1/4096$ at index 0 gives strictness when $A > 0$.

For R , relabel the support by $p \mapsto 4 - p$. Its index polynomial becomes

$$C(z) = 1 + \frac{1}{4}z^2 + \frac{1}{4}z^3 + \frac{9}{64}z^4.$$

The coefficients of $C(z)^4$ at indices 0 through 7 are

$$1, \quad 0, \quad 1, \quad 1, \quad \frac{15}{16}, \quad \frac{3}{4}, \quad \frac{55}{64}, \quad \frac{39}{64}.$$

For indices at least 8, write $C = 1 + H$. In $C^4 = 1 + 4H + 6H^2 + 4H^3 + H^4$, the first two terms do not contribute. The contribution of $6H^2$ is at most $3/8$ and occurs only at index 8. Since H is coefficientwise bounded by $z^2(1 + z + z^2)/4$, the other two contributions are at most $7/16$ and $19/256$. Here 7 and 19 are the largest coefficients of $(1 + z + z^2)^3$ and $(1 + z + z^2)^4$. The total is at most $227/256 < 1$. Thus all coefficients pass, and the coefficient $15/16$ at reflected index 4 gives strictness for positive kernels. Reflection reverses the output indices and preserves their sum. $\square$

Lemma 7.5. *For $k \geq 0$, $(k \cdot V_L)^4 \leq T_F(k)$, strictly when k is positive on F .*

Proof. The linear form is $E = A + B/4 + 9C/64 + D/4$. The coefficient sums in E^4 are listed below. An omitted entry is zero.

$$\begin{array}{c}
 \begin{array}{cc}
 n & c_n \\
 \hline
 0 & 1 \\
 1 & 1 \\
 2 & 15/16 \\
 3 & 31/64 \\
 4 & 1 + 467/2048 \\
 5 & 3/4 + 279/4096 \\
 6 & 39/64 + 1215/65536 \\
 7 & 29/128 + 729/262144 \\
 8 & 3/8 + 351/4096 + 6561/16777216 \\
 \hline
 \end{array}
 \begin{array}{cc}
 n & c_n \\
 \hline
 9 & 3/16 + 243/16384 \\
 10 & 33/256 + 729/262144 \\
 11 & 27/1024 \\
 12 & 1/16 + 243/32768 \\
 13 & 1/64 \\
 14 & 9/1024 \\
 15 & 0 \\
 16 & 1/256 \\
 \hline
 \end{array}
 \end{array} \tag{7.20}$$

Only sum 4 exceeds one. Its polynomial is

$$A^3D + \frac{243}{2048}A^2C^2 + \frac{27}{256}AB^2C + \frac{1}{256}B^4 \leq U + aV + bW,$$

where

$$\begin{aligned}
 U &= A^3D, \quad V = A^2C^2, \quad W = B^4, \\
 a &= \frac{351}{2048}, \quad b = \frac{29}{512}, \quad a + b = \frac{467}{2048}.
 \end{aligned} \tag{7.21}$$

This uses $AB^2C \leq (A^2C^2 + B^4)/2$. For $M = \max(U, V, W)$,

$$U + aV + bW \leq M + a \min(U, V) + b \min(U, W). \quad (7.22)$$

Indeed, $a(V - U)_+ + b(W - U)_+ \leq (a + b)(M - U) \leq M - U$. Weighted AM–GM also gives

$$\begin{aligned} \min(A^3D, A^2C^2) &\leq \frac{4}{15}A^3C + \frac{25}{12}A^2D^2, \\ \min(A^3D, B^4) &\leq \frac{4}{15}A^2B^2 + \frac{25}{12}A^2D^2. \end{aligned} \quad (7.23)$$

For the first line, the minimum is at most $U^{2/3}V^{1/3}$, which equals $(A^3C)^{2/3}(A^2D^2)^{1/3}$. The stated bound follows since $(4/15)^2(25/12) = 4/27$. The second line has the same exponents.

The term M is at most the maximum at sum 4. All added terms from (7.23) go to sums 2 and 8. The final coefficient sum at 2 is

$$\frac{15}{16} + \frac{467}{2048} \frac{4}{15} = \frac{7667}{7680} < 1. \quad (7.24)$$

The original coefficient sum at 8 is less than $15/32$, and the added weight is

$$\frac{467}{2048} \frac{25}{12} = \frac{11675}{24576} < \frac{1}{2}.$$

Its final total is below $31/32$. The remaining sums pass by (7.20). For positive kernels, the positive maximum and remaining slack at sum 2 give strictness. $\square$

Lemma 7.6. *For $k \geq 0$, $(k \cdot V_R)^4 \leq T_F(k)$, strictly when k is positive on F .*

Proof. Relabel the support by $p \mapsto 2 - p$, and label the kernel values on positions $(0, 1, 2, -2)$ by (A, B, C, D) . The profile is then $(1, 1/4, 9/64, 1/4)$, so the target is again $E = A + B/4 + 9C/64 + D/4$. Only sums $0, 1, 2, -2$ have coefficient sum greater than one. Retain respectively

$$A^4, \quad A^3B, \quad \frac{9}{16}A^3C + \frac{3}{8}A^2B^2, \quad A^3D. \quad (7.25)$$

The terms removed from these groups are

sum	removed terms	
0	$(3/16)AB^2D + (27/64)A^2CD + (243/32768)C^2D^2$	
1	$(1/64)B^3D + (27/128)ABCD$	
2	$(27/1024)B^2CD + (243/4096)AC^2D$	
-2	$(3/128)B^2D^2 + (27/256)ACD^2$	

(7.26)

Apply the following weighted AM–GM inequalities:

$$\begin{aligned} AB^2D &\leq (A^2D^2 + B^4)/2, & A^2CD &\leq (A^2C^2 + A^2D^2)/2, \\ C^2D^2 &\leq (C^4 + D^4)/2, & B^3D &\leq (3B^4 + D^4)/4, \\ ABCD &\leq (A^2D^2 + B^2C^2)/2, & B^2CD &\leq B^4/2 + C^4/4 + D^4/4, \\ AC^2D &\leq (A^2D^2 + C^4)/2, & B^2D^2 &\leq (B^4 + D^4)/2, \\ ACD^2 &\leq (A^2C^2 + D^4)/2. \end{aligned} \quad (7.27)$$

Every destination is a fourfold product on the reflected support. Only sums $4, -4, 6, 8, -8$ receive additions. Their coefficient sums are

sum	original	addition	final	
4	$467/2048 + 729/262144$	$807/2048$	$163801/262144 < 5/8$	
-4	$3/8 + 9/1024$	$3603/8192$	$6747/8192 < 7/8$	
6	$1215/65536$	$27/256$	$8127/65536 < 1/8$	
8	$6561/16777216$	$2619/65536$	$677025/16777216 < 1/16$	
-8	$1/256$	$5155/65536$	$5411/65536 < 3/32$	

(7.28)

Besides these and the four groups in (7.25), the unchanged nonzero coefficient sums are

sum	coefficient sum	sum	coefficient sum
-6	1/16	3	31/64 + 243/16384
-5	1/64	5	279/4096
-3	3/16	7	729/262144
-1	3/4 + 27/1024		

(7.29)

All are less than one. The retained groups at 0, 1, -2 have coefficient sum one, and the group at 2 has coefficient sum 15/16. This proves the inequality and gives strictness for positive kernels at sum 2. Reflection preserves the sum of the fourfold maxima. $\square$

Proposition 7.7. *For $F = \{0, 1, 2, 4\}$ and every nonnegative kernel k ,*

$$N_F(k)^4 \leq T_F(k). \quad (7.30)$$

The inequality is strict if k is positive on F .

Proof. Lemma 7.3 bounds the weighted objective by its values on (7.2) and on feasible profiles with a zero coordinate. Lemmas 7.1, 7.4, 7.5, and 7.6 bound the fourth power of each of these values by $T_F(k)$.

For positive kernels every listed comparison is strict. The feasible set is compact by (7.1), so its weighted maximum is attained. Applying the strict comparison to a maximizing profile proves the strict form of (7.30). $\square$

7.4 The sharp unweighted norm

The same reduction determines the unweighted norm exactly.

Proposition 7.8. *For $F = \{0, 1, 2, 4\}$,*

$$\sup_{\substack{0 \neq x \geq 0 \\ \text{supp}(x) \subseteq F}} \frac{\|x\|_2^2}{\|x * x\|_\infty} = \frac{105}{64}. \quad (7.31)$$

Equality holds for $x = (3/8, 1/2, 1/2, 1)$ in the coordinate order $(0, 1, 2, 4)$.

Proof. By homogeneity, normalize $\|x * x\|_\infty = 1$. In Lemma 7.3, the totals of V_L, V_R, R are 105/64, those of H_L, H_R are 13/8 = 104/64, and that of H_0 is 3/2. Every feasible profile with a zero coordinate has total at most 3/2 by Lemma 7.1. This gives the upper bound.

The squared profile of the stated x is R . It satisfies (7.1): the two progression left sides are 5/8 and 1, all pair bounds hold, and the coordinate at 4 is 1. Thus $\|x * x\|_\infty = 1$ and $\|x\|_2^2 = 105/64$. $\square$

8 The support-size boundary

Proposition 8.1. *For every nonempty integer set F with $|F| \leq 3$ and every nonnegative kernel k supported on F , one has $N_F(k)^4 \leq T_F(k)$.*

Proof. Sets with at most two points are Sidon, as is any three-point set that is not an arithmetic progression. Proposition 3.2 applies to all of them. For a three-term progression, complete it to a four-term progression and extend both x and k by zero. This preserves ordinary convolution, the weighted objective, and $T_F(k)$. The four-term-progression inequality proved above therefore gives the desired bound for every feasible x , and then for $N_F(k)$. $\square$

For a singleton $F = \{p\}$ the bound is equality: $N_F(k) = k(p)$ and $T_F(k) = k(p)^4$. The corresponding max-convolution inequality is also equality for every nonzero f, g allowed in Corollary 1.3, since

$$\|f \star g\|_1 = g(p)\|f\|_1, \quad \|f \star \widetilde{gk}\|_1 = g(p)k(p)\|f\|_1.$$

Thus the strict four-point statement does not extend to singleton supports.

The preceding proposition, the classification in Proposition 2.1, and the five four-point cases prove the positive part of Theorem 1.1. The following example proves its support-size assertion.

Proposition 8.2. *Let $F = \{0, 2, 7, 8, 11\}$ and $k = 1_F$. Then*

$$N_F(k) = \frac{5}{2}, \quad T_F(k) = 39, \quad N_F(k)^4 - T_F(k) = \frac{1}{16}.$$

Proof. The fifteen unordered pair sums, including diagonal pairs, are

$$0, 2, 4, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18, 19, 22.$$

They are distinct, so F is Sidon. For $x = 1_F/\sqrt{2}$, ordinary convolution equals $1/2$ at diagonal sums and 1 at off-diagonal sums. Consequently $x * x \leq 1$ and $\sum_{p \in F} x(p)^2 = 5/2$. Equation (3.2), which holds for Sidon sets of any size, also supplies the upper bound: $N_F(1_F) = \max\{5/2, (5+3)/4\} = 5/2$.

To count $4F$, put $E = \{0, 2, 7, 8\}$ and write $[r, s]_{\mathbb{Z}} = \{r, r+1, \dots, s\}$ for integer endpoints $r \leq s$. For $h \geq 0$, partition the summands in hE by the number m drawn from $\{7, 8\}$. Their sum ranges from $7m$ to $8m$ in unit steps, while the remaining summands from $\{0, 2\}$ contribute $2j$ for $0 \leq j \leq h-m$. Hence

$$hE = \bigcup_{m=0}^h \bigcup_{j=0}^{h-m} [7m+2j, 8m+2j]_{\mathbb{Z}}. \quad (8.1)$$

In particular,

$$4E = \{0, 2, 4\} \cup [6, 26]_{\mathbb{Z}} \cup [28, 32]_{\mathbb{Z}}.$$

Now partition $4F$ by the number ℓ of summands equal to 11 . Formula (8.1) gives the new elements at each step:

Layer	Elements not in preceding layers	Number
$4E$	$\{0, 2, 4\} \cup [6, 26]_{\mathbb{Z}} \cup [28, 32]_{\mathbb{Z}}$	29
$11 + 3E$	$\{27, 33, 34, 35\}$	4
$22 + 2E$	$\{36, 37, 38\}$	3
$33 + E$	$\{40, 41\}$	2
$\{44\}$	$\{44\}$	1

Thus

$$4F = \{0, 2, 4\} \cup [6, 38]_{\mathbb{Z}} \cup \{40, 41, 44\}, \quad |4F| = 39.$$

The fourfold max-convolution of an indicator is the indicator of its fourfold sumset, so $T_F(1_F) = 39$. Finally, $(5/2)^4 = 625/16 = 39 + 1/16$. $\square$

The example shows failure of the sufficient covering bound $N_F(k)^4 \leq T_F(k)$ with both x and k positive throughout a five-point set. A counterexample to the reflected max-convolution inequality would additionally require functions f, g for which the ratio in (1.5) exceeds $T_F(k)^{1/4}$; this construction supplies no such functions.

9 Sharp unweighted covering constants

The five pair-sum patterns have different unweighted norms. Their values, with one maximizing squared profile in the natural coordinate order, are listed below.

Four-point pattern	$N_F(1_F)$	Maximizing squared profile
Sidon	2	$(1/2, 1/2, 1/2, 1/2)$
Isolated progression, $(0, a, 2a, b)$	$7/4$	$(1/4, 1/4, 1/4, 1)$
Two progressions, $(0, 1, 2, 4)$	$105/64$	$(9/64, 1/4, 1/4, 1)$
Parallelogram, $(0, a, b, a + b)$	$25/16$	$(1, 1/4, 1/4, 1/16)$
Four-term progression	$3/2$	$(1, 1/4, 0, 1/4)$

For Sidon supports, (3.2) gives $\max\{2, 7/4\} = 2$. The isolated-progression profile sums are $105/64$ twice, $3/2$ four times, and $7/4$ twice. The sixteen parallelogram profile sums are $3/2$ for the twelve profiles supported on three corners and $25/16$ for the four tensor profiles. For a four-term progression, the five profile sums before reversal are $3/2, 3/2, 381/256, 3/2, 89/64$. The double-progression reduction bounds every profile by ones with total at most $105/64$, and its displayed profile is feasible. Thus the reductions prove all five upper bounds, while the listed feasible profiles attain them.

By Lemma 1.2, each value is also the maximum of $L_F(g, 1_F)$ over nonzero nonnegative g , and the supremum when g is required to be positive at every point. This sharpness concerns the universal translated covering problem. The max-convolution ratio in (1.5) can be smaller than its covering bound.

10 Reflected ratios and general kernels

We first bound arbitrary kernels and give lower examples with the same leading coefficient. Exact constants through four sites and averaging bounds for larger supports then yield fourfold consequences.

For finite nonempty $H \subset \mathbb{Z}$ and nonzero finitely supported $f, g \geq 0$, write

$$P = \|f \star g\|_1, \quad R_H(f, g) = \frac{\|f \star \widetilde{g 1_H}\|_1}{P}.$$

The denominator includes all values of g , including those outside H .

10.1 An energy bound for arbitrary kernels

The equality of sum and difference representation energies is classical; Hennecart, Robert and Yudin [2, p. 174, equation (4)] record it for equal inputs. We use its mixed weighted form and a pointwise quadratic estimate to bound the reflected ratio for every kernel.

Theorem 10.1 (Weighted energy bound). *Let $k \geq 0$ be finitely supported on $\mathbb{Z}$, and put $K = \|k\|_1$, $M = \|k\|_\infty$. For every nonzero finitely supported $f, g \geq 0$,*

$$\frac{\|f \star \widetilde{gk}\|_1}{\|f \star g\|_1} \leq \frac{(\sqrt{K} + \sqrt{M})^2}{4}. \quad (10.1)$$

The profile g may vanish on the kernel support and may have values outside it; the denominator includes all of g .

Proof. The zero kernel is immediate. Otherwise enumerate its support as $F = \{p_1, \dots, p_n\}$, with $k_i = k(p_i) > 0$. Put

$$d = \sqrt{KM}, \quad t = K + d, \quad q(x) = t \sum_i k_i x_i^2 - \left(\sum_i k_i x_i \right)^2.$$

We first prove, for every nonnegative vector x ,

$$K \max_i k_i x_i^2 \leq q(x) \leq \frac{t^2}{4} \max_i x_i^2. \quad (10.2)$$

For the upper bound, normalize $\max_i x_i = 1$ and set $S = \sum_i k_i x_i$. Then $\sum_i k_i x_i^2 \leq S$, so $q(x) \leq tS - S^2 \leq t^2/4$. Homogeneity gives the general case, and the zero vector is immediate.

For the lower bound fix i , set $a = x_i$, $W = K - k_i$, and $S_i = \sum_{j \neq i} k_j x_j$. When $W > 0$, weighted Cauchy–Schwarz gives

$$\begin{aligned} q(x) &\geq t \left(k_i a^2 + \frac{S_i^2}{W} \right) - (k_i a + S_i)^2 \\ &= \frac{t - K + k_i}{W} \left(S_i - \frac{k_i W a}{t - K + k_i} \right)^2 + k_i a^2 \frac{t(t - K)}{t - K + k_i}. \end{aligned}$$

All denominators are positive. Since $k_i \leq M$ and $d^2 = KM$,

$$\frac{t(t - K)}{t - K + k_i} \geq \frac{(K + d)d}{d + M} = K.$$

When $W = 0$, the kernel has one point, $K = M = k_i$, $t = 2K$, and $q(x) = K^2 x_i^2$ directly. Thus $q(x) \geq K k_i x_i^2$ for every i , which proves the lower bound in (10.2).

Now write $g_i = g(p_i)$ and define

$$x_i^+(s) = \sqrt{g_i f(s - p_i)}, \quad x_i^-(s) = \sqrt{g_i f(s + p_i)}.$$

Their weighted square sums have the same total:

$$\sum_s \sum_i k_i (x_i^+(s))^2 = \sum_s \sum_i k_i (x_i^-(s))^2 = \|f\|_1 \sum_i k_i g_i.$$

Set $h = \sqrt{f}$ and $w(p_i) = k_i \sqrt{g_i}$, with $w = 0$ outside F . Then $\sum_i k_i x_i^+ = (h * w)$ and $\sum_i k_i x_i^- = (h * \tilde{w})$. The mixed energy identity is

$$\|h * w\|_2^2 = \|h * \tilde{w}\|_2^2. \quad (10.3)$$

Indeed, exchanging the two w coordinates turns the relation $a + p = b + q$ into $a - q = b - p$ without changing the product of the four weights. Consequently $\sum_s q(x^+(s)) = \sum_s q(x^-(s))$.

Put $Q = \|f \star \widetilde{gk}\|_1$ and $P = \|f \star g\|_1 > 0$. The reflected weighted squared maxima sum to Q ; the forward unweighted squared maxima sum to $P_F = \|f \star (g1_F)\|_1 \leq P$. Summing (10.2) gives

$$KQ \leq \sum_s q(x^-(s)) = \sum_s q(x^+(s)) \leq \frac{t^2}{4} P_F \leq \frac{t^2}{4} P.$$

Division by KP proves (10.1), since $t^2/K = (\sqrt{K} + \sqrt{M})^2$. $\square$

Corollary 10.2. *Let $F \subset \mathbb{Z}$ have $n \geq 1$ elements. For every nonzero finitely supported $f, g \geq 0$,*

$$R_F(f, g) = \frac{\|f \star \widetilde{g} 1_F\|_1}{\|f \star g\|_1} \leq C_n, \quad C_n = \frac{(\sqrt{n} + 1)^2}{4}. \quad (10.4)$$

The inequality is strict when $n > 1$. No support restriction is imposed on g outside F .

Proof. Apply Theorem 10.1 with $k = 1_F$. For strictness, let $F_+ = \{p \in F : g(p) > 0\}$ and $m = |F_+|$. If $m = 0$ the numerator vanishes; if $m = 1$ the singleton bound gives $R_F(f, g) \leq 1 < C_n$ for $n > 1$.

When $m > 1$, apply the preceding proof with $k = 1_{F_+}$ and $t = m + \sqrt{m}$. Put $a = \min \text{supp}(f)$, $b = \max F_+$, and $s_0 = a - b$. Exactly one coordinate of $x^-(s_0)$ is nonzero: its square is $g(b)f(a) > 0$. Hence the lower pointwise comparison has the strict gap

$$q(x^-(s_0)) - m \max_i (x_i^-(s_0))^2 = (\sqrt{m} - 1)g(b)f(a) > 0.$$

All other gaps are nonnegative. Summing therefore gives $R_F(f, g) < C_m \leq C_n$, still with the full denominator P . $\square$

For a singleton, equality in (10.4) is possible. Theorem 10.9 gives the sharp four-site bound. Corollary 10.13 gives sharper bounds for five and six points; the latter proves the constant-kernel fourfold inequality when $|4F| \geq 48$. It does not settle every six-point support.

Remark 10.3 (Kernel level sets). A bound retaining the distribution of the kernel values follows from Corollary 10.2. For nonzero k , let $E_u = \{p : k(p) > u\}$ and $N(u) = |E_u|$. Keep f, g and $P = \|f \star g\|_1$ fixed. Subadditivity in the kernel gives

$$\frac{\|f \star \widetilde{g} k\|_1}{P} \leq \int_0^M C_{N(u)} du = \frac{K + M}{4} + \frac{1}{2} \int_0^M \sqrt{N(u)} du.$$

Indeed, at each output the maximum of the kernel integrals is at most the integral of the maxima, and $\int_0^M N(u) du = K$. For descending positive kernel values $k_1 \geq \dots \geq k_r > 0$, with $k_{r+1} = 0$, the integral is the finite sum $\sum_{j=1}^r (k_j - k_{j+1})C_j$; tied values contribute zero. Cauchy–Schwarz gives $\int_0^M \sqrt{N(u)} du \leq \sqrt{MK}$, recovering (10.1). Only the kernel varies across levels; the profile and its full denominator remain fixed.

10.2 Lower examples and the leading coefficient

Colesanti’s orthant exponential supplies an earlier template for this product construction [5, Theorems 1.1–1.2]. For a nonnegative log-concave h on $\mathbb{R}^d$ with $0 < \int h^2 < \infty$, his difference-function inequality gives $\|h \star \widetilde{h}\|_1 / \|h \star h\|_1 \leq 2^d$; here $\star$ uses a supremum and the norms use Lebesgue measure. Geometric sequences also occur in the max-convolution examples of Matolcsi, Ruzsa, Shakan and Zhelezov [6, Example 11.2]; related reflected product-exponential examples appear in Madiman, Manui, Zawalski and Zvavitch [13, Lemma A.3]. The following adaptation samples and truncates the exponential product, restricts the second profile to the simplex vertices, and counts the boundary faces. It gives a lower bound for prescribed kernel values when their integer locations may vary.

Proposition 10.4. *Let $d \geq 1$, $w_0 = M > 0$ and $0 \leq w_i \leq M$ for $1 \leq i \leq d$, and put $K = \sum_{i=0}^d w_i$. There are finite integer examples with these kernel values whose reflected ratios tend to $(K + 3M)/4$. In particular, for $n \geq 2$,*

$$\sup_{\substack{|F|=n \\ f, g \neq 0}} R_F(f, g) \geq \frac{n + 3}{4},$$

where the supremum ranges over integer supports and nonnegative finitely supported inputs.

Proof. Work first in $\mathbb{Z}^d$. For an integer $N \geq 1$, let

$$A_N = \{0, \dots, N\}^d, \quad H = \{0, e_1, \dots, e_d\}, \quad f_N(x) = 2^{-\sum_i x_i} 1_{A_N}(x).$$

Set $g(0) = 1$, $g(e_i) = 1/2$, $k(0) = M$, $k(e_i) = w_i$, and let g, k vanish elsewhere. Write $S_N = \sum_{j=0}^N 2^{-j} = 2 - 2^{-N}$.

For $y \in A_N$, the zero-support contribution to $(f_N \star g)(y)$ is $f_N(y)$, and every available other contribution is also $\frac{1}{2}f_N(y - e_i) = f_N(y)$. Outside A_N , positive values occur exactly on the disjoint faces $y_i = N + 1$, $0 \leq y_j \leq N$ for $j \neq i$. Their values are $2^{-N-1}2^{-\sum_{j \neq i} y_j}$. Thus the full denominator is

$$P_N = S_N^d + d2^{-N-1}S_N^{d-1}.$$

For the reflected numerator, the contribution at zero is $Mf_N(y)$ on A_N , and the others are $\frac{1}{2}w_i f_N(y + e_i) = \frac{1}{4}w_i f_N(y) \leq Mf_N(y)$ when available. Exterior contributions lie on the disjoint negative faces $y_i = -1$, $0 \leq y_j \leq N$ for $j \neq i$, with value $\frac{1}{2}w_i 2^{-\sum_{j \neq i} y_j}$. Therefore

$$Q_N = MS_N^d + \frac{K - M}{2}S_N^{d-1}, \quad \frac{Q_N}{P_N} = \frac{MS_N + (K - M)/2}{S_N + d2^{-N-1}} \rightarrow \frac{K + 3M}{4}.$$

For integer examples put $B_N = 2N + 3$ and $\phi_N(x) = \sum_{i=1}^d x_i B_N^{i-1}$. This additive map is injective on $\{-1, \dots, N + 1\}^d$. Indeed, at the highest nonzero coordinate of a difference, the term has magnitude at least B_N^j , while all lower terms have total magnitude at most

$$(N + 2) \frac{B_N^j - 1}{B_N - 1} < B_N^j.$$

It preserves the input values and all collisions in $A_N + H$ and $A_N - H$. Pushing f_N, g, k forward by ϕ_N and extending them by zero therefore preserves both norms. The kernel locations are $\{0, 1, B_N, \dots, B_N^{d-1}\}$; for $d > 1$ these vary with N . Finally take $w_i = 1$ for every i , so that $K = d + 1 = n$ and $M = 1$. $\square$

For indicator inputs, a standard lattice-simplex construction also gives the same asymptotic leading coefficient. The counts below are the specialization $(d, l_1, l_2) = (n - 1, n - 2, 1)$ of Glasscock's Lemma B.1 [3, pp. 37–38]. That source develops the simplex family of Hennecart, Robert and Yudin [2, Section 3].

Proposition 10.5. *For every integer $n \geq 3$, there are a set $F \subset \mathbb{Z}$ with n elements and a finite nonempty set $A' \subset \mathbb{Z}$ such that*

$$R_F(1_{A'}, 1_F) = \frac{|A' - F|}{|A' + F|} = \frac{n^2 - 2}{4n - 6}. \quad (10.5)$$

Proof. Put $d = n - 1$, $m = n - 2$, and in $\mathbb{Z}^d$ let $U = \{0, e_1, \dots, e_d\}$ and $A = mU$, with repetitions allowed. Thus A consists of the nonnegative integer vectors whose coordinate sum is at most m . The simplex counts give

$$|A + U| = \binom{m + n}{n - 1}, \quad |A - U| = \binom{m + n - 1}{n - 1} + (n - 1) \binom{m + n - 2}{n - 2}.$$

For the difference count, the nonnegative points are precisely A . Each remaining point has exactly one coordinate equal to -1 ; after choosing that coordinate, its other nonnegative

coordinates sum to at most m . The classes for distinct negative coordinates are disjoint. With $m = n - 2$, the quotient simplifies to

$$\frac{|A - U|}{|A + U|} = \frac{1}{2} + \frac{(n-1)^2}{2(2n-3)} = \frac{n^2-2}{4n-6}.$$

To obtain integer sets, choose $B = m + 3$ and use the additive encoding $\phi(x) = \sum_{j=1}^d x_j B^{j-1}$. Every coordinate of every point in $(A + U) \cup (A - U)$ lies between -1 and $m + 1$. Adding the same all-ones vector puts the coordinates in $\{0, \dots, m + 2\}$, strictly below B . Uniqueness of base- B expansions proves injectivity on that union. Consequently $F = \phi(U)$ has n points, and $A' = \phi(A)$ satisfies (10.5). $\square$

Corollary 10.6. *For $n \geq 1$, let K_n be the supremum of $R_F(f, g)$ over all n -point sets $F \subset \mathbb{Z}$ and all nonzero finitely supported $f, g \geq 0$. Then $K_1 = 1$, and*

$$\frac{n+3}{4} \leq K_n \leq \frac{(\sqrt{n}+1)^2}{4} \quad (n \geq 2), \quad \lim_{n \rightarrow \infty} \frac{K_n}{n} = \frac{1}{4}.$$

Proof. The singleton case is immediate. Proposition 10.4 gives the lower bound, and Corollary 10.2 gives the general upper bound. Dividing by n proves the limit. $\square$

The geometric lower bound exceeds the displayed simplex ratio:

$$\frac{n+3}{4} - \frac{n^2-2}{4n-6} = \frac{3n-5}{8n-12} > 0 \quad (n \geq 3).$$

The first construction permits weighted inputs; the second uses indicators. Pointwise strictness of the upper bounds gives non-strict bounds on K_n . The leading coefficient $1/4$ is asymptotically optimal; no optimality is asserted for the finite remainder when $n \geq 5$.

10.3 Allocation bounds and exact constants

Let $F = \{p_1, \dots, p_n\}$ support a nonzero kernel k , and write $k_i = k(p_i) > 0$, $g_i = g(p_i)$ and $I = \{i : g_i > 0\}$. For each positive output of $f \star \widetilde{gk}$, select a maximizing site. Let E_i be the disjoint sets of outputs selected for p_i , and put

$$a_i = \frac{g_i}{P} \sum_{y \in E_i} f(y + p_i) \quad (i \in I), \quad a_i = 0 \quad (i \notin I).$$

The winners are chosen for the weighted numerator, so $\|f \star \widetilde{gk}\|_1 / P = \sum_i k_i a_i$. Since $P \geq g_i \|f\|_1$ and each selected sum is at most $\|f\|_1$, we have $0 \leq a_i \leq 1$. For distinct $i, j \in I$, translating the disjoint sets E_i, E_j by the same $p_i + p_j$ gives

$$P \geq g_j \sum_{y \in E_i} f(y + p_i) + g_i \sum_{y \in E_j} f(y + p_j).$$

Division by P and AM–GM yield

$$\frac{g_j}{g_i} a_i + \frac{g_i}{g_j} a_j \leq 1, \quad a_i a_j \leq \frac{1}{4}. \quad (10.6)$$

The product bound holds for every distinct pair, including zero allocations, without dividing by a zero weight.

Theorem 10.7 (Weighted allocation bound). *For every nonnegative finitely supported kernel k and nonzero finitely supported $f, g \geq 0$, put $K = \|k\|_1$ and $M = \|k\|_\infty$. Then*

$$\frac{\|f \star \widetilde{gk}\|_1}{\|f \star g\|_1} \leq \max \left\{ \frac{K}{2}, \frac{K + 3M}{4} \right\}. \quad (10.7)$$

There is no restriction on the number or locations of the kernel sites.

Proof. The zero kernel or zero restricted profile gives zero. Otherwise use the weighted winner allocations above. If every $a_i \leq 1/2$, then $\sum_i k_i a_i \leq K/2$. If $x = a_j > 1/2$, the pair product bound gives $a_i \leq 1/(4x)$ for every $i \neq j$, and therefore

$$\sum_i k_i a_i \leq k_j x + \frac{K - k_j}{4x}.$$

This expression is convex on $[1/2, 1]$. Its endpoint values are $K/2$ and $(K + 3k_j)/4 \leq (K + 3M)/4$, proving the result. $\square$

Corollary 10.8 (Sharpness for $K \leq 3M$). *For any prescribed nonempty list of positive kernel values with sum K and maximum M satisfying $K \leq 3M$, the supremum of the reflected ratio over all distinct integer locations and all nonzero finite nonnegative inputs is exactly*

$$\frac{K + 3M}{4}. \quad (10.8)$$

In particular, this applies to every kernel on at most three sites, and

$$K_1 = 1, \quad K_2 = \frac{5}{4}, \quad K_3 = \frac{3}{2}.$$

Proof. When $K \leq 3M$, the second term in (10.7) is the larger one. Proposition 10.4 gives the matching lower bound for each list with at least two values; a singleton attains the value M by translation. Taking all values equal to one proves the three scalar identities. Integer locations may vary in the lower construction. No attainment on every fixed support is asserted. $\square$

10.4 The sharp four-site bound

Theorem 10.9 (Sharp four-site ratio). *For every four-element $H \subset \mathbb{Z}$ and all nonzero finitely supported $f, g \geq 0$,*

$$R_H(f, g) = \frac{\|f \star \widetilde{g1_H}\|_1}{\|f \star g\|_1} \leq \frac{7}{4}. \quad (10.9)$$

The denominator retains all values of g , including those outside H . The supremum over the inputs and the four integer sites is $K_4 = 7/4$.

The upper bound follows from an inequality for four events and finite linear-programming duality. We first prove the probability inequality.

Lemma 10.10. *Let $m = (a, b, c, d)$ with $a \geq b \geq c \geq d > 0$ and $d/a \geq 1/3$. Define $r_{ij} = \min(m_i, m_j) / \max(m_i, m_j)$ for $0 \leq i, j \leq 3$, and let $\mathcal{R} = (r_{ij})$. Then, as quadratic forms,*

$$\mathcal{R} \succeq \frac{5}{8} \mathbf{11}^T.$$

Proof. Set $q_1 = b/a$, $q_2 = c/b$, $q_3 = d/c$, so $0 < q_i \leq 1$ and $q_1 q_2 q_3 \geq 1/3$. For an orthonormal basis e_0, e_1, e_2, e_3 , define

$$v_0 = e_0, \quad v_i = q_i v_{i-1} + \sqrt{1 - q_i^2} e_i.$$

Their Gram matrix is $\mathcal{R}$. Set

$$u = e_0 + \sum_{i=1}^3 \sqrt{\frac{1 - q_i}{1 + q_i}} e_i.$$

Induction gives $u \cdot v_i = 1$ for every i , including when $q_i = 1$. If $t_i = (1 - q_i)/(1 + q_i)$ and $S = t_1 + t_2 + t_3$, then $\|u\|^2 = 1 + S$.

To bound S , put $P = t_1 t_2 + t_1 t_3 + t_2 t_3$ and $T = t_1 t_2 t_3$. The product condition gives

$$3(1 - S + P - T) \geq 1 + S + P + T,$$

$$2S \leq 1 + P - 2T \leq 1 + \frac{S^2}{3}.$$

Thus $S^2 - 6S + 3 \geq 0$. As $0 \leq S < 3$, this forces $S < 3/5$: on $[3/5, 3]$ that polynomial is decreasing and its value at $3/5$ is $-6/25$. Hence $\|u\|^2 < 8/5$. For any real vector ξ ,

$$\left(\sum_i \xi_i \right)^2 = \left(u \cdot \sum_i \xi_i v_i \right)^2 \leq \frac{8}{5} \left\| \sum_i \xi_i v_i \right\|^2 = \frac{8}{5} \xi^T \mathcal{R} \xi.$$

This proves the claimed non-strict bound. $\square$

Lemma 10.11 (Four-event star inequality). *Let J be a random subset of $\{0, 1, 2, 3\}$, with*

$$m_i = \mathbb{P}(i \in J) > 0, \quad c_{ij} = \mathbb{P}(i, j \in J), \quad r_{ij} = \frac{\min(m_i, m_j)}{\max(m_i, m_j)}.$$

Put

$$T = \sum_i m_i, \quad D_i = \sum_{j \neq i} r_{ij} c_{ij}, \quad W = \max_i D_i.$$

Then

$$T - W \leq \frac{7}{4}.$$

Consequently the same bound holds when W is replaced by the maximum weight of a spanning tree, with edge weights $r_{ij} c_{ij}$. The constant $7/4$ is attained for both versions.

Proof. Reductions.

We may relabel so that

$$1 \geq a = m_0 \geq b = m_1 \geq c = m_2 \geq d = m_3 > 0.$$

The union bound gives $c_{0j} \geq a + m_j - 1$, even when the right side is negative. Therefore

$$T - D_0 \leq a + \frac{b(1 - b) + c(1 - c) + d(1 - d)}{a} \leq a + \frac{3}{4a}. \quad (10.10)$$

For $3/4 \leq a \leq 1$ the last quantity is at most $7/4$, since

$$a + \frac{3}{4a} - \frac{7}{4} = \frac{(4a - 3)(a - 1)}{4a} \leq 0.$$

For any three positive-marginal events, their analogous maximum-star deficit is at most $3/2$. Indeed, if their largest marginal u is less than $1/2$, their total marginal is less than $3/2$. Otherwise the same union-bound argument gives a deficit at most

$$u + \frac{1}{2u} \leq \frac{3}{2}, \quad \frac{1}{2} \leq u \leq 1,$$

because the difference is $(2u - 1)(u - 1)/(2u)$. Deleting the event with marginal d and retaining a maximizing three-event star thus proves the claimed bound when $d \leq 1/4$.

Let

$$A = \sum_{i < j} r_{ij} c_{ij}, \quad N = |J|.$$

Always $W \geq A/2$, by averaging the four star degrees. If $d > 1/4$ and $a \leq 1/2$, all ratios are at least $1/2$, so

$$A \geq \frac{1}{2} \mathbb{E} \binom{N}{2} \geq \frac{T-1}{2}.$$

The second inequality follows from $\binom{k}{2} \geq k - 1$ for integers $0 \leq k \leq 4$. As $T \leq 2$, we obtain

$$T - W \leq T - \frac{T-1}{4} = \frac{3T+1}{4} \leq \frac{7}{4}.$$

The case $T \leq 7/4$ is also immediate from nonnegative edge weights.

It remains to consider

$$\frac{1}{2} < a < \frac{3}{4}, \quad d > \frac{1}{4}, \quad T > \frac{7}{4}. \quad (10.11)$$

If the first bound in (10.10) is at most $7/4$, there is nothing more to prove. Otherwise, write

$$a = \frac{1}{2} + x, \quad b = \frac{1}{2} + y, \quad c = \frac{1}{2} + z, \quad d = \frac{1}{2} + w, \quad s = y + z + w, \quad E = y^2 + z^2 + w^2.$$

The failure of that bound implies

$$0 < x < \frac{1}{4}, \quad E < R_0 := \left(\frac{1}{2} - x\right) \left(\frac{1}{4} - x\right) = x^2 - \frac{3}{4}x + \frac{1}{8}. \quad (10.12)$$

Indeed the numerator $b(1-b) + c(1-c) + d(1-d)$ is $3/4 - E$.

A variance estimate.

Let $Y_i = 1_{\{i \in J\}}$ and $m = (a, b, c, d)^T$. Under (10.11), $d/a > 1/3$, so Lemma 10.10 applies to $Y - m$ pointwise. Taking expectations yields

$$\mathbb{E}[Y^T \mathcal{R} Y] \geq m^T \mathcal{R} m + \frac{5}{8} \text{Var } N.$$

The left side is $T + 2A$, and the ordering of the marginals gives

$$m^T \mathcal{R} m = a^2 + 3b^2 + 5c^2 + 7d^2.$$

Consequently

$$T - W \leq \frac{5T - (a^2 + 3b^2 + 5c^2 + 7d^2) - (5/8) \text{Var } N}{4}. \quad (10.13)$$

It is enough to prove

$$D + \frac{5}{8} \text{Var } N \geq 0, \quad D = a^2 + 3b^2 + 5c^2 + 7d^2 - 5T + 7. \quad (10.14)$$

In the variables of (10.12), $T = 2 + x + s$ and direct expansion gives

$$D = 1 - 4x + x^2 - 2y + 2w + 3y^2 + 5z^2 + 7w^2. \quad (10.15)$$

The case $T \leq 2$.

Here $s \leq -x$. We prove the stronger assertion $D > 0$. Completing squares and applying weighted Cauchy–Schwarz gives

$$\begin{aligned} D &= 1 - 4x + x^2 - \frac{10}{21} + 3(y - 1/3)^2 + 5z^2 + 7(w + 1/7)^2 \\ &\geq 1 - 4x + x^2 - \frac{10}{21} + \frac{105}{71}(s - 4/21)^2 \\ &\geq \frac{176x^2 - 244x + 41}{71}. \end{aligned}$$

For $0 < x \leq 3/16$, the numerator is decreasing and is at least $23/16$, so this is positive.

Suppose instead $x \geq 3/16$. The inequalities $E \geq s^2/3 \geq x^2/3$ and $E < R_0$ imply

$$R_0 - \frac{x^2}{3} = \frac{2}{3}x^2 - \frac{3}{4}x + \frac{1}{8} > 0.$$

This quadratic is decreasing on $[0, 1/4]$ and has value $-1/1936$ at $9/44$, so $x < 9/44$.

Orthogonal projection onto the plane perpendicular to $(1, 1, 1)$ gives

$$(y - w)^2 \leq 2(E - s^2/3) \leq 2(E - x^2/3).$$

Using $3y^2 + 5z^2 + 7w^2 \geq 3E$ in (10.15), we obtain

$$D \geq 1 - 4x + x^2 + 3E - 2\sqrt{2(E - x^2/3)}.$$

As a function of E on the allowed interval, the last two terms are decreasing: their derivative is $3 - 2/\sqrt{2(E - x^2/3)} < 0$, since $E \leq R_0 \leq 1/8$. (The lower endpoint follows by continuity.)

Thus

$$D \geq \frac{11}{8} - \frac{25x}{4} + 4x^2 - 2\sqrt{2(R_0 - x^2/3)}.$$

For $3/16 \leq x < 9/44$, the decreasing function $R_0 - x^2/3$ is at most $1/128$, its value at $3/16$. Also $11/8 - 25x/4 + 4x^2$ is decreasing here and is at least $511/1936$, its value at $9/44$. Hence

$$D \geq \frac{511}{1936} - \frac{1}{4} = \frac{27}{1936} > 0.$$

This proves (10.14) throughout the case $T \leq 2$.

The case $T \geq 2$.

Since $a < 3/4$, we have $2 \leq T < 3$. For every integer N the product $(N - 2)(N - 3)$ is nonnegative, so

$$\text{Var } N \geq (T - 2)(3 - T).$$

Put $t = x + s = T - 2$, $\rho = 5/8$, and $h = \rho(1 - 2x)$. By expansion,

$$\begin{aligned} D + \rho t(1 - t) &= B + (-2 + h)y + hz + (2 + h)w \\ &\quad + 3y^2 + 5z^2 + 7w^2 - \rho s^2, \\ B &= 1 - \frac{27x}{8} + \frac{3x^2}{8}. \end{aligned}$$

Weighted Cauchy–Schwarz gives

$$s^2 \leq \frac{23}{15}(y^2 + 3z^2 + 5w^2), \quad \rho \frac{23}{15} = \frac{23}{24} < 1.$$

It follows that

$$3y^2 + 5z^2 + 7w^2 - \rho s^2 \geq 2E.$$

The squared length of the linear coefficient vector is

$$L^2 = (-2 + h)^2 + h^2 + (2 + h)^2 = 8 + \frac{75}{64}(1 - 2x)^2.$$

Therefore

$$\begin{aligned} D + \rho t(1 - t) &\geq B - L\sqrt{E} + 2E \\ &\geq B - L\sqrt{R_0} + 2R_0 \\ &= Z - L\sqrt{R_0}, \end{aligned}$$

where $Z = (10 - 39x + 19x^2)/8$. The function $2E - L\sqrt{E}$ is decreasing for $0 \leq E \leq R_0 \leq 1/8$, as follows either by differentiation or by setting $E = v^2$ and noting $4v - L < 0$. Also $Z \geq 23/128 > 0$ for $0 \leq x \leq 1/4$.

For completeness, the remaining rational polynomial check is displayed explicitly:

$$\begin{aligned} Z^2 - L^2 R_0 &= \frac{F(4x)}{16384}, \\ F(u) &= 61u^4 - 3828u^3 + 16824u^2 - 19344u + 6816. \end{aligned} \tag{10.16}$$

Let $B_{4,k}(v) = \binom{4}{k} v^k (1 - v)^{4-k}$. Direct expansion gives the following Bernstein coefficients:

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$
$F(v/2)$	6816	4398	2681	12363/8	14005/16
$F((1 + v)/2)$	14005/16	821/4	3/4	146	529

The table means that each polynomial in the left column equals the sum of the listed coefficient times $B_{4,k}(v)$. All coefficients are positive. As $B_{4,k}(v) \geq 0$ and $\sum_k B_{4,k}(v) = 1$ on $[0, 1]$, we have $F(u) > 0$ on $[0, 1]$. Equation (10.16), together with $Z > 0$, gives $Z > L\sqrt{R_0}$. This proves (10.14) for $T \geq 2$.

Together the reductions and the two cases prove the star inequality. Every star is a spanning tree, so the tree inequality follows.

Equality.

Give probability $1/6$ to each of

$$\{0, 1\}, \quad \{0, 2\}, \quad \{0, 3\}, \quad \{0, 1, 2\}, \quad \{0, 1, 3\}, \quad \{0, 2, 3\}.$$

Then $m_0 = 1$ and $m_1 = m_2 = m_3 = 1/2$, while $c_{0j} = 1/2$ and $c_{ij} = 1/6$ for distinct leaves. Thus every center–leaf edge has weight $1/4$, and every leaf–leaf edge has weight $1/6$. The center star has weight $3/4$; each leaf star has weight $7/12$. Every spanning tree has three edges, each of weight at most $1/4$, and the center star attains that bound. As $T = 5/2$, both deficits are exactly $7/4$. $\square$

Proof of Theorem 10.9. If g vanishes at some point of H , remove its zero coordinates. Corollary 10.8 bounds the resulting ratio by $3/2$; if all coordinates vanish, the ratio is zero. Thus suppose $H = \{p_0, p_1, p_2, p_3\}$ and $g_i = g(p_i) > 0$ for every i . Put $G = \sum_i g_i$ and $C = 7/4$, and define

$$\rho_{ij} = \frac{\min(g_i, g_j)}{\max(g_i, g_j)}, \quad \kappa_{ij} = \frac{\min(g_i, g_j)}{g_i + g_j},$$

$$\Phi_{ij}(v_i, v_j) = \min(v_i, v_j) + \min\left(\frac{g_j}{g_i}v_i, \frac{g_i}{g_j}v_j\right).$$

Bounding both minima by their j arguments gives

$$\kappa_{ij}\Phi_{ij}(v_i, v_j) \leq \frac{\min(g_i, g_j)}{g_j}v_j \leq v_j,$$

and the analogous bound holds with i . For any star on the four indices, root its edges at an index maximizing v_i and charge each edge to its child. Every other coordinate is charged once. Consequently, if $w_i \geq 0$ and $\sum_i w_i = 1$ are probabilities for the four star centers, then

$$\begin{aligned} W_w(v) &= \sum_{i < j} (w_i + w_j) \kappa_{ij} \Phi_{ij}(v_i, v_j), \\ q_-(v) &= \sum_i v_i - W_w(v) \geq \max_i v_i. \end{aligned}$$

We seek real diagonal coefficients d_i with

$$\sum_i g_i d_i = G \tag{10.17}$$

such that $q_+(v) = \sum_i d_i v_i - W_w(v) \leq C \max_i v_i$. The function q_+ is convex and positively homogeneous. Its upper bound therefore follows from its values on the unit-cube vertices:

$$\sum_{i \in J} d_i - \sum_{\{i,j\} \subset J} (w_i + w_j) \rho_{ij} \leq C \quad (\emptyset \neq J \subseteq \{0, 1, 2, 3\}). \tag{10.18}$$

Here $\kappa_{ij}\Phi_{ij}(1, 1) = \rho_{ij}$; the empty vertex is zero.

Maximize $\sum_i g_i d_i$ subject to (10.18), $w_i \geq 0$ and $\sum_i w_i = 1$. This finite linear program is feasible with $d_i = 0$. The singleton constraints give $d_i \leq C$. Since every $g_i > 0$, a maximizing level set can also be bounded below in every d_i , so the maximum is finite and attained. Its dual value is

$$\min_{\lambda} \left[C \sum_{J \neq \emptyset} \lambda_J + \max_i \sum_{j \neq i} \rho_{ij} \sum_{J \supset \{i,j\}} \lambda_J \right], \tag{10.19}$$

where $\lambda_J \geq 0$ is indexed by the nonempty subsets and $\sum_{J \ni i} \lambda_J = g_i$ for every i . Indeed, the multipliers of (10.18) must have the displayed marginals because the d_i are unrestricted. Their remaining objective is linear in w , so its maximum on the probability simplex is the largest of the four star sums.

For each feasible λ , set $L = \sum_J \lambda_J > 0$ and give the nonempty subset J probability λ_J/L . Its marginals are $m_i = g_i/L$ and its pair marginals are $c_{ij} = L^{-1} \sum_{J \supset \{i,j\}} \lambda_J$. The ratios of the m_i equal those of the g_i . Lemma 10.11 shows that the expression in (10.19) is at least $L \sum_i m_i = G$. Finite linear-programming duality therefore supplies w, d with objective at least G . Decreasing all d_i by a common nonnegative constant preserves (10.18) and gives the exact trace (10.17).

Finally set $v_i^\pm(z) = g_i f(z \mp p_i)$. Translation by $p_i + p_j$ interchanges the two terms of $\Phi_{ij}(v_i^+, v_j^+)$ with those of $\Phi_{ij}(v_i^-, v_j^-)$, so their sums over z agree. Also $\sum_z v_i^\pm(z) = g_i \|f\|_1$. The trace condition and the two pointwise comparisons now give

$$\begin{aligned} \|f \star \widetilde{g1_H}\|_1 &\leq \sum_z q_-(v^-(z)) = \sum_z q_+(v^+(z)) \\ &\leq C \|f \star (g1_H)\|_1 \leq C \|f \star g\|_1. \end{aligned}$$

This proves the upper bound with the full denominator. Proposition 10.4, with four kernel values equal to one, gives the matching lower bound on the supremum. $\square$

Corollary 10.12 (Sharp weighted bound on four sites). *Let $k \geq 0$ have at most four positive sites, and put $K = \|k\|_1$ and $M = \|k\|_\infty$. For all nonzero finitely supported $f, g \geq 0$,*

$$\frac{\|f \star \widetilde{gk}\|_1}{\|f \star g\|_1} \leq \frac{K + 3M}{4}. \quad (10.20)$$

For every prescribed nonempty list of at most four positive kernel values, the supremum over the inputs and distinct integer locations is exactly $(K + 3M)/4$.

Proof. The zero kernel is immediate. Otherwise let $F_t = \{p : k(p) > t\}$ for $0 \leq t < M$, and keep $P = \|f \star g\|_1$ fixed. Every F_t is nonempty and has at most four elements. At each output, the maximum of the kernel integrals is at most the integral of the maxima. Thus Corollary 10.8 and Theorem 10.9 give

$$\begin{aligned} \|f \star \widetilde{gk}\|_1 &\leq \int_0^M \|f \star \widetilde{g1_{F_t}}\|_1 dt \\ &\leq \frac{P}{4} \int_0^M (|F_t| + 3) dt = \frac{K + 3M}{4} P. \end{aligned}$$

The sum over outputs is finite, and each application uses the same profile and full denominator. Proposition 10.4 gives sharpness for each list of at least two values; a singleton attains the value M . The lower construction permits the integer locations to vary. It does not assert attainment on every fixed support. $\square$

Corollary 10.13 (Averaging four-subsets). *For every n -element integer set H , $n \geq 4$, and all nonzero finitely supported $f, g \geq 0$,*

$$R_H(f, g) \leq \frac{7n}{16}. \quad (10.21)$$

In particular,

$$K_5 \leq \frac{35}{16}, \quad K_6 \leq \frac{21}{8}.$$

For a positive constant kernel, the reflected fourfold inequality is strict on five-point supports with $|4H| \geq 23$ and six-point supports with $|4H| \geq 48$.

Proof. For each output z , choose a maximizing site $p \in H$ in $\max_{q \in H} g(q)f(z + q)$. Every four-subset containing p has the same maximum, and there are $\binom{n-1}{3}$ such subsets. Hence

$$\max_{q \in H} g(q)f(z + q) \leq \frac{1}{\binom{n-1}{3}} \sum_{\substack{J \subset H \\ |J|=4}} \max_{q \in J} g(q)f(z + q).$$

This also holds for zero maxima and ties. Summing over z and applying Theorem 10.9 to each J , with the same full denominator $P = \|f \star g\|_1$, gives

$$R_H(f, g) \leq \frac{\binom{n}{4}}{\binom{n-1}{3}} \frac{7}{4} = \frac{7n}{16}.$$

For $k = c1_H$, $c > 0$, the reflected numerator is $c\|f \star \widetilde{g1_H}\|_1$ and $\|k \star k \star k \star k\|_1 = c^4|4H|$. The fourfold conclusions follow from the strict comparisons

$$35^4 = 1500625 < 1507328 = 23 \cdot 16^4, \quad 21^4 = 194481 < 196608 = 48 \cdot 8^4.$$

$\square$

The ratio bounds in (10.21) are non-strict. Their fourth-power comparisons give the stated strict fourfold conclusions. The six-point conclusion concerns constant kernels and requires $|4H| \geq 48$.

10.5 Five-point supports

The reflected fourfold inequality extends beyond the support-size boundary of the covering bound when the kernel is constant.

Theorem 10.14. *Let $F \subset \mathbb{Z}$ have five elements. For every nonzero finitely supported $f, g \geq 0$,*

$$\|f \star \widetilde{g1_F}\|_1^4 < |4F| \|f \star g\|_1^4.$$

In particular, the five-point set in Proposition 8.2 satisfies this inequality despite failing the sufficient covering bound.

For an ordinary-autoconvolution-feasible $x \geq 0$, write $w_p = x(p)^2$. The universal constraints are $0 \leq w_p \leq 1$ and $w_p w_q \leq 1/4$ for distinct points. We use repeatedly the elementary inequality

$$b + c \leq a + \frac{bc}{a} \quad (0 \leq b, c \leq a, a > 0), \quad (10.22)$$

which follows from $(a - b)(a - c) \geq 0$.

Lemma 10.15. *If a five-element integer set F contains four distinct points forming a parallelogram, then $N_F(1_F) \leq 2$.*

Proof. Label the four points so that $p_0 + p_3 = p_1 + p_2$, and label the fifth point p_4 . The extra convolution constraint is

$$\sqrt{w_0 w_3} + \sqrt{w_1 w_2} \leq \frac{1}{2}.$$

If all weights are at most $1/2$, apply (10.22) to the two opposite pairs. With $p = \sqrt{w_0 w_3}$ and $q = \sqrt{w_1 w_2}$,

$$w_0 + w_1 + w_2 + w_3 \leq 1 + 2(p^2 + q^2) \leq 1 + 2(p + q)^2 \leq \frac{3}{2}.$$

Adding $w_4 \leq 1/2$ proves the bound in this region.

Otherwise let $u \in [1/2, 1]$ be a largest weight, and put $a = 1/(4u) \in [1/4, 1/2]$. Every other weight is at most a . If $u = w_4$, then

$$\sum_i w_i \leq u + 2a + \frac{p^2 + q^2}{a}, \quad 0 \leq p, q \leq a, \quad p + q \leq \frac{1}{2}.$$

On this polygon the maximum of $p^2 + q^2$ is $a^2 + (1/2 - a)^2$, as follows by checking its vertices. Hence

$$\sum_i w_i \leq 2u + \frac{1}{u} - 1 \leq 2.$$

The last function is convex and equals two at both endpoints.

If the largest weight is a corner, relabel so that $u = w_0$ and put $p = \sqrt{w_1 w_2} \in [0, a]$. The opposite weight is at most $(1/2 - p)^2/u$. Equation (10.22) and $w_4 \leq a$ give

$$\sum_i w_i \leq u + 2a + \frac{p^2}{a} + \frac{(1/2 - p)^2}{u}.$$

This is convex in p , so the endpoint bounds are

$$A(u) = u + \frac{3}{4u}, \quad B(u) = u + \frac{1}{u} - \frac{1}{4u^2} + \frac{1}{16u^3}.$$

Both are convex on $[1/2, 1]$; for the second, $B''(u) = (2u^2 - 3u/2 + 3/4)/u^5 > 0$. Their endpoint values are $(2, 7/4)$ and $(2, 29/16)$, respectively. Thus every feasible profile has total at most two. $\square$

Lemma 10.16. *If a five-element integer set F contains a three-term progression, then $N_F(1_F) \leq 17/8$.*

Proof. Write b, c for the endpoint weights and v for the middle weight. The extra constraint is

$$v + 2\sqrt{bc} \leq 1. \quad (10.23)$$

If every weight is at most $1/2$, then (10.22) gives

$$b + c + v \leq \frac{1}{2} + 2bc + v \leq \frac{1}{2} + \frac{(1-v)^2}{2} + v = 1 + \frac{v^2}{2} \leq \frac{9}{8}.$$

The other two weights contribute at most one.

Otherwise let $u \in [1/2, 1]$ be a largest weight and set $a = 1/(4u)$. All remaining weights are at most a . There are three cases.

An endpoint is largest. Take $b = u$. For $v \in [0, a]$, the other endpoint is at most $(1-v)^2/(4u) \leq a$, so

$$\sum w \leq u + 2a + v + \frac{(1-v)^2}{4u}.$$

Convexity in v reduces this to $v = 0, a$, giving $A(u)$ above and

$$C(u) = u + \frac{1}{u} - \frac{1}{8u^2} + \frac{1}{64u^3}.$$

The function C is convex because $C''(u) = (2u^2 - 3u/4 + 3/16)/u^5 > 0$. Its endpoint values are $17/8$ and $121/64$; also $A(u) \leq 2$.

The middle point is largest. Take $v = u$. Equations (10.22) and (10.23) give $b + c \leq a + u(1-u)^2$. Thus

$$\sum w \leq D(u) = u + \frac{3}{4u} + u(1-u)^2.$$

Here $D''(u) = 6u - 4 + 3/(2u^3) > 0$ on $[1/2, 1]$, and its endpoint values are $17/8$ and $7/4$.

An outside point is largest. The other outside weight is at most a , and

$$b + c + v \leq v + a + \min \left\{ a, \frac{(1-v)^2}{4a} \right\}, \quad 0 \leq v \leq a.$$

If $u \in [3/4, 1]$, the simple bound $b + c + v \leq 3a$ gives $\sum w \leq u + 1/u \leq 25/12 < 17/8$. If $u \in [1/2, 3/4]$, then $a \in [1/3, 1/2]$. The two branches meet at $v_0 = 1 - 2a \in [0, a]$. The first branch increases to one. The second is convex, with endpoint values one and $2a + (1-a)^2/(4a)$. Therefore

$$\sum w \leq \max \left\{ u + \frac{1}{4u} + 1, 2u + \frac{13}{16u} - \frac{1}{2} \right\}.$$

Both functions are convex on $[1/2, 3/4]$. Their endpoint values are $(2, 25/12)$ and $(17/8, 25/12)$, respectively. This completes the cases. $\square$

Proposition 10.17. *If a five-element integer set F is not Sidon, then $N_F(1_F)^4 < |4F|$.*

Proof. If F contains a four-point parallelogram, Lemma 10.15 gives $N_F(1_F)^4 \leq 16 < 17 \leq |4F|$. The last bound follows from repeated application of $|A + B| \geq |A| + |B| - 1$ in $\mathbb{Z}$.

Otherwise every pair-sum collision consists of one diagonal pair and one off-diagonal pair, forming a three-term progression. A sum cannot have two different off-diagonal pairs, since they would be disjoint and form a parallelogram. Each point is the centre of at most one progression, and the two extreme points are never centres. Thus there are at most three collisions among the fifteen unordered pair sums, giving

$$|2F| \geq 12, \quad |4F| \geq 2|2F| - 1 \geq 23.$$

Lemma 10.16 now gives $N_F(1_F)^4 \leq 83521/4096 < 23$. $\square$

Lemma 10.18 (An elementary Sidon bound). *Every five-element Sidon subset $F \subset \mathbb{Z}$ satisfies $|4F| \geq 30$.*

Proof. We first recall the equality case of $|A + A| \geq 2|A| - 1$. Translate a finite nonempty A so that $\min A = 0$, and put $L = \max A$. The singleton case is immediate. For $L > 0$, $A \cup (L + A) \subseteq A + A$ has $2|A| - 1$ elements, since the two sets meet only at L . Equality therefore forces $A + A = A \cup (L + A)$.

The image B of A in $\mathbb{Z}/L\mathbb{Z}$ is then closed under addition and contains zero. It is a subgroup: for every $b \in B$, some $r \geq 1$ has $rb = 0$, so $-b = (r - 1)b \in B$. If $B = \{0\}$, put $d = L$. Otherwise its least positive representative d divides L , since the remainder on dividing L by d belongs to B and is smaller than d . The same argument shows that every element of B is a multiple of d , and closure supplies all multiples. Because $A \subset [0, L]$ and both endpoints belong to A , it follows that $A = \{0, d, \dots, L\}$. Conversely, every arithmetic progression has $|A + A| = 2|A| - 1$.

Now translate F so that $\min F = 0$. The Sidon property gives $|2F| = 15$, hence $|4F| \geq 29$. If equality held, the characterization just proved would give $2F = \{0, d, \dots, 14d\}$. Since $F \subset 2F$ and $2 \max F = \max(2F) = 14d$, we would have $F \subset \{0, d, \dots, 7d\}$. Its ten positive differences are distinct: $x - y = u - v > 0$ gives $x + v = u + y$, so the Sidon property forces the same ordered pair. They cannot all lie in the seven-element set $\{d, \dots, 7d\}$. Thus $|4F| \neq 29$, proving the assertion. $\square$

Proof of Theorem 10.14. The non-Sidon case follows from Proposition 10.17 and Lemma 1.2. For Sidon supports, combine Lemma 10.18 with Corollary 10.13, using $(35/16)^4 < 23 \leq 30$. $\square$

For $f = 1_A$ and $g = 1_F$, Theorem 10.14 gives $|A - F|^4 < |A + F|^4 |4F|$ for every finite nonempty integer set A and every five-element integer set F . The theorem also permits arbitrary nonnegative weights in f and g ; the kernel k is fixed to 1_F . For later use, we record an estimate for constant denominator weights. For a set J of $s \geq 1$ integers, put

$$e_s = \frac{1}{s} \left\lfloor \frac{(s+1)^2}{4} \right\rfloor.$$

Then

$$\|f \star 1_{-J}\|_1 \leq e_s \|f \star 1_J\|_1.$$

Indeed, for a finite nonempty set B , the ordinary sum and difference representation functions have the same squared energy $\mathcal{E}$, by exchanging the two J coordinates in the corresponding quadruples. Their common total mass is $W = s|B|$, and their positive values are integers r from one to s . The inequalities

$$(s+1)r - \left\lfloor \frac{(s+1)^2}{4} \right\rfloor \leq r^2 \leq (s+1)r - s$$

give

$$(s+1)W - \left\lfloor \frac{(s+1)^2}{4} \right\rfloor |B+J| \leq \mathcal{E} \leq (s+1)W - s|B-J|.$$

Hence $|B-J| \leq e_s|B+J|$. Integration over the level sets of f proves the stated max-convolution estimate. In particular, for five-element F ,

$$\|f \star 1_{-F}\|_1 \leq \frac{9}{5} \|f \star 1_F\|_1. \quad (10.24)$$

If $g > 0$ on F and $r = \max_F g / \min_F g$, comparison with the indicator kernels gives, still with the full denominator,

$$R_F(f, g) \leq \frac{9}{5} r. \quad (10.25)$$

The next subsection records the stronger sharp Sidon cardinality bound.

10.6 Five-point Sidon supports

Proposition 10.19. *Every five-element Sidon subset F of $\mathbb{Z}$ satisfies $|4F| \geq 39$. Equality is attained by $F = \{0, 2, 7, 8, 11\}$.*

Proof. We use Freiman's classical $3k-4$ theorem in the following form: if $A \subset \mathbb{Z}$, $|A| = k \geq 3$ and $|A+A| \leq 3k-4$, then A is contained in an arithmetic progression with at most $|A+A| - k + 1$ elements. This is the equal-set specialization of the statement recalled by Bollobás, Leader and Tiba [11, p. 1].

Translate F and divide by the greatest common divisor of its differences, writing

$$F = \{0, a, b, c, d\}, \quad 0 < a < b < c < d, \quad \gcd(F) = 1.$$

These operations preserve the Sidon property and sumset cardinalities. Set $B = F+F$. Then $|B| = 15$, $\min B = 0$, $\max B = 2d$, and $\gcd(B) = 1$ because $F \subset B$. If $|4F| = |B+B| \leq 38$, the theorem applies, since $38 \leq 3 \cdot 15 - 4$. It puts B in an arithmetic progression with at most $38 - 15 + 1 = 24$ elements. Its step divides every difference of B , so its absolute value is one. Consequently $2d \leq 23$, and $d \leq 11$.

The ten positive differences of a five-point Sidon set are distinct, so $d \geq 10$. If $d = 10$, those differences would be $1, \dots, 10$, whose sum is odd. Their sum is instead $4d + 2c - 2a$, which is even. Thus the supposition $|4F| \leq 38$ forces $d = 11$.

We classify this last case by hand. The ten differences omit one number m from $1, \dots, 11$, and the parity argument shows that m is even. Put $P_i = \{i, 11-i\}$ for $1 \leq i \leq 5$. The six endpoint differences

$$a, b, c, 11-a, 11-b, 11-c$$

occupy exactly three distinct pairs P_i . The three remaining differences $b-a, c-b, c-a$ form a Schur triple: the sum of the smaller two is the largest. They must be three of the four members of the two unused pairs, say P_r, P_s with $r < s$.

Deleting one element of the ordered list $r, s, 11-s, 11-r$ must therefore leave a Schur triple. Deleting the smallest cannot work, since $s + (11-s) = 11 > 11-r$. The other deletions give $s = 2r$, $2r + s = 11$, or $r + 2s = 11$. Their complete solution list is

Omitted m	Unused (r, s)	Interior differences	$c - a$
2	(1, 2)	$\{1, 9, 10\}$	10
4	(2, 4)	$\{2, 7, 9\}$	9
6	(3, 5)	$\{3, 5, 8\}$	8
8	(3, 4)	$\{3, 4, 7\}$	7
10	(1, 5)	$\{1, 5, 6\}$	6

The interior points must occupy the other three complementary pairs, once each. For $m = 2$, the span $c - a = 10$ is impossible inside $[1, 10]$. For $m = 4$, it forces $(a, c) = (1, 10)$, using the same complementary pair. For $m = 6$, the possibilities $(a, c) = (1, 9), (2, 10)$ give only $b = 4, 7$, respectively. For $m = 8$, the possible endpoint pairs are $(1, 8), (2, 9), (3, 10)$; the outer two use the forbidden P_3 , and the middle one repeats P_2 . For $m = 10$, the choices $a = 1, 4$ use the forbidden P_1 . The remaining choices $(a, c) = (2, 8), (3, 9)$ give only $b = 7, 4$, respectively. Up to reflection, the only sets are therefore

$$F_1 = \{0, 1, 4, 9, 11\}, \quad F_2 = \{0, 2, 7, 8, 11\}.$$

For the first representative,

$$2F_1 = \{0, 1, 2, 4, 5, 8, 9, 10, 11, 12, 13, 15, 18, 20, 22\}.$$

Successive elements are at most three apart, so $2F_1 + \{0, 1, 2\}$ contains $[0, 24]$. The rest of $[0, 38]$ is supplied by

$$\begin{aligned} [25, 28] &= 15 + [10, 13], & [29, 33] &= 20 + [9, 13], \\ [34, 35] &= 22 + [12, 13], & 36 &= 18 + 18, \\ 37 &= 15 + 22, & 38 &= 18 + 20. \end{aligned}$$

All summands belong to $2F_1$. Also $40 = 18 + 22$, $42 = 20 + 22$ and $44 = 22 + 22$. The only other candidates are $39, 41, 43$; each requires a summand 20 or 22 , but its complementary summand is absent. Hence

$$4F_1 = [0, 38] \cup \{40, 42, 44\}, \quad |4F_1| = 42.$$

For the second representative, Proposition 8.2 gives

$$4F_2 = \{0, 2, 4\} \cup [6, 38] \cup \{40, 41, 44\}, \quad |4F_2| = 39.$$

Here all intervals are integer intervals. Both possibilities contradict $|4F| \leq 38$, and the second proves sharpness. $\square$

The proof classifies normalized span 11 only. It does not classify all equality cases for $|4F| = 39$.

10.7 Kernels on sets with distinct four-term sums

Call a finite set F a B_4 set if every equality $p_1 + p_2 + p_3 + p_4 = q_1 + q_2 + q_3 + q_4$, with all entries in F , has the same multiset of entries on both sides. Thus its unordered four-term sums are distinct, with repetition allowed. This is the $h = 4$ support condition in Ruzsa's independent addition graph [4, Chapter 1, Section 5, pp. 11–12].

Corollary 10.20. *Let $F \subset \mathbb{Z}$ have $n \geq 1$ elements. Suppose that F is a B_4 set or that $|4F| \geq ((\sqrt{n} + 1)^2/4)^4$. For a constant kernel $k = c1_F$, $c \geq 0$, and every nonzero finitely supported $f, g \geq 0$,*

$$\|f \star \widetilde{gk}\|_1^4 \leq \|f \star g\|_1^4 \|k \star k \star k \star k\|_1.$$

The inequality is strict when $n > 1$ and $c > 0$.

Proof. For $k = c1_F$, the two kernel-dependent quantities are

$$\|f \star \widetilde{gk}\|_1 = c\|f \star \widetilde{g1_F}\|_1, \quad \|k \star k \star k \star k\|_1 = c^4|4F|.$$

Corollary 10.2 proves the assertion under the fourfold-cardinality hypothesis. When $c = 0$ both sides vanish.

For a B_4 set, $|4F| = \binom{n+3}{4}$. Write $x = \sqrt{n} \geq 1$. Direct expansion gives

$$768 \left[\binom{n+3}{4} - C_n^4 \right] = (x-1)(29x^7 + 5x^6 + x^4(113x - 55) + x^2(87x - 81) + 27x + 3). \quad (10.26)$$

The bracket is positive for $x \geq 1$. Thus $|4F| \geq C_n^4$, with strict inequality for $n > 1$, and the first part applies. $\square$

Corollary 10.21. *Let $k \geq 0$ be a nonzero kernel supported on a finite B_4 set, with $K = \|k\|_1$, $M = \|k\|_\infty$, and $K/M \geq 9$. For every nonzero finitely supported $f, g \geq 0$,*

$$\|f \star \widetilde{gk}\|_1^4 < \|f \star g\|_1^4 \|k \star k \star k \star k\|_1.$$

No support restriction is imposed on f or g .

Proof. Enumerate the positive values of k as $k_1, \dots, k_r$. The B_4 condition gives

$$\|k \star k \star k \star k\|_1 = h_4(k) := \sum_{\substack{\alpha_1, \dots, \alpha_r \geq 0 \\ \alpha_1 + \dots + \alpha_r = 4}} \prod_i k_i^{\alpha_i}.$$

Each represented sum has one multiset of summands, whose product is its max-convolution value. The multinomial expansion therefore gives

$$K^4 = \sum_{|\alpha|=4} \frac{4!}{\prod_i \alpha_i!} \prod_i k_i^{\alpha_i} \leq 24h_4(k).$$

Since $K/M \geq 9$ and $24 \cdot 4^4 = 6144 < 6561 = 9^4$, Theorem 10.1 gives

$$\frac{\|f \star \widetilde{gk}\|_1}{\|f \star g\|_1} \leq \frac{K}{4} (1 + \sqrt{M/K})^2 \leq \frac{4K}{9} < \frac{K}{24^{1/4}} \leq h_4(k)^{1/4}.$$

Taking fourth powers proves the result. $\square$

The weighted corollary requires both the B_4 condition and $K/M \geq 9$; it includes non-constant kernels. It does not establish the fourfold inequality for all kernel weights on B_4 sets or on arbitrary supports.

11 Arbitrary kernels on a five-point Sidon support

The support that defeats the universal covering bound also satisfies the reflected inequality for every choice of kernel weights. The proof uses rational coefficient certificates in addition to the estimates below.

Theorem 11.1. *Let $F = \{0, 2, 7, 8, 11\}$. Every nonnegative kernel k supported on F and every pair of nonnegative finitely supported functions f, g on $\mathbb{Z}$ satisfy*

$$\|f \star \widetilde{gk}\|_1^4 \leq T_F(k) \|f \star g\|_1^4.$$

The inequality is strict when k is positive at all five points and f, g are nonzero.

This result concerns the displayed support. Theorem 10.14 allows every five-point integer support when k is constant. Neither theorem assumes a bound on the size or diameter of the support of f .

11.1 Allocations from a common input

Initially suppose that $f \neq 0$ and that g is positive on a five-point set $F = \{p_1, \dots, p_5\}$ and zero elsewhere. Write $P = \|f \star g\|_1 > 0$. For each positive term of

$$N = \|f \star \widetilde{gk}\|_1 = \sum_y \max_i k_i g_i f(y + p_i),$$

select a maximizing index, resolving ties by index order. Let E_i be the disjoint sets of outputs assigned to index i , and put

$$a_i = \frac{g_i}{P} \sum_{y \in E_i} f(y + p_i).$$

Then $N/P = \sum_i k_i a_i$. Denominator outputs at $y + 2p_i$ give $0 \leq a_i \leq 1$. For distinct i, j , translate both E_i and E_j by $p_i + p_j$. They remain disjoint, and the corresponding denominator witnesses give

$$\frac{g_j}{g_i} a_i + \frac{g_i}{g_j} a_j \leq 1, \quad a_i a_j \leq \frac{1}{4}. \quad (11.1)$$

Thus these constraints hold simultaneously for allocations obtained from the same f, g , even though their selection depends on k .

Lemma 11.2. *Every allocation above satisfies*

$$\sum_i a_i \leq c := \frac{1109}{450}.$$

Proof. Put $R = \max_i g_i / \min_i g_i$ and $t = 1/R$. We first show that

$$A := \sum_i a_i \leq F(t) := 2 + t - \frac{t^2}{2}. \quad (11.2)$$

The pair at the largest and smallest values of g satisfies $ta + b/t \leq 1$. If $a, b \leq v \leq 1$, filling the coordinate with the smaller coefficient first gives

$$a + b \leq \min\{2v, v + t - vt^2\}.$$

When $M := \max_i a_i \leq 1/2$, take $v = 1/2$ and bound the other three coordinates by $1/2$ to obtain (11.2).

For $1/2 \leq M \leq 1$, all other coordinates are at most $1/(4M)$ by (11.1). If the maximum is outside the extremal pair, the preceding two-variable bound gives

$$A \leq \min\{U(M), V(M)\}, \quad U(M) = M + \frac{1}{M}, \quad V(M) = M + \frac{3 - t^2}{4M} + t.$$

If the maximum belongs to the pair, assigning it to the smaller coefficient can only increase the allowed partner. This gives the same first bound and the second bound $M + 3/(4M) + t - Mt^2 \leq V(M)$.

The functions switch order at $M_0 = (1 + t^2)/(4t) \geq 1/2$. The function U decreases on $[1/2, 1]$, while V is convex there. Consequently it suffices to check $V(1/2) = F(t)$, $V(1) \leq F(t)$, and, if $M_0 \leq 1$, $U(M_0) \leq F(t)$. The endpoint difference is $F(t) - V(1) = (1 - t^2)/4$. At the switch,

$$4t(1 + t^2)(F(t) - U(M_0)) = (1 - t)^2(-2t^3 - t^2 + 6t - 1).$$

Here $M_0 \leq 1$ implies $t \geq 2 - \sqrt{3} > 1/4$. On $[1/4, 1]$, the last factor is at least $-3t^2 + 6t - 1 = 2 - 3(1 - t)^2 \geq 5/16$. This proves (11.2).

At each output, the selected unweighted product is at most $\max_i g_i f(y + p_i)$. Therefore the energy estimate (10.25) also gives $A \leq 9R/5$. Combining the two bounds yields

$$A \leq \min \left\{ \frac{9R}{5}, 2 + \frac{1}{R} - \frac{1}{2R^2} \right\}.$$

For $R \leq 15/11$, the first expression is at most $27/11 < c$. For $R \geq 15/11$, the second is decreasing and is at most c . $\square$

For the kernel, write $S = \sum_i k_i$, $K = \max_i k_i$, and $m = \min_i k_i$. Define

$$\delta = \frac{5}{2} - c = \frac{8}{225}, \quad \alpha = \frac{c - \sqrt{c^2 - 4}}{2}, \quad \beta = \frac{1}{4\alpha}.$$

Then $1/2 < \alpha < 1$ and $\alpha + 4\beta = c$.

Lemma 11.3. *The allocations satisfy*

$$\sum_i k_i a_i \leq \max\{U_c, V_c, C\}, \quad U_c = \frac{S}{2} - \delta m, \quad V_c = \alpha K + \beta(S - K), \quad C = \frac{S + 3K}{4}.$$

Proof. Use only $0 \leq a_i \leq 1$, $a_i a_j \leq 1/4$, and $\sum_i a_i \leq c$. These constraints are invariant under coordinate permutations, so rearrangement permits a largest allocation M at a largest kernel weight. If $M \leq 1/2$, the deficit from five halves is at least δ ; its weighted cost is at least δm , giving U_c .

If $1/2 \leq M \leq \alpha$, the other four coordinates are at most $1/(4M)$. Since $M + 1/M \geq c$, subtracting the least possible cost of the missing mass bounds the objective by

$$mc + (K - m)M + \frac{S - K - 4m}{4M}.$$

The coefficient $S - K - 4m$ is nonnegative, so this expression is convex in M . Its endpoint values are U_c and V_c . For $\alpha \leq M \leq 1$, the convex bound $KM + (S - K)/(4M)$ has endpoint values V_c and C . $\square$

11.2 The corner comparison

Lemma 11.4. *If F is a five-point Sidon set with $|4F| \geq 32$, then*

$$T_F(k) \geq \left(\frac{S + 3K}{4} \right)^4.$$

The inequality is strict when at least two kernel coordinates are positive.

Proof. The zero kernel is immediate. Normalize $K = 1$, choose a position p_i with weight one, and let $u_1, \dots, u_4$ be the other weights. Set $t = \sum_j u_j \in [0, 4]$. Sidonicity makes the 15 outputs $2p_i + (p_j + p_\ell)$, $j \leq \ell$, distinct. Their total contribution is at least

$$1 + t + \frac{t^2 + \sum_j u_j^2}{2} \geq 1 + t + \frac{5t^2}{8}.$$

At least 17 other fourfold outputs each contribute at least m^4 , where $m \geq (t - 3)_+$. Hence

$$T_F(k) - (1 + t/4)^4 \geq t^2 A(t) + 17(t - 3)_+^4, \quad A(t) = \frac{1}{4} - \frac{t}{16} - \frac{t^2}{256}.$$

The function A decreases on $[0, 4]$. For $0 < t \leq 13/4$ it is at least $23/4096 > 0$. On the remaining intervals, the displayed lower bound is bounded below respectively by

$$\begin{array}{l|l} [13/4, 10/3] & -25/1296 + 17/256 = 977/20736 \\ [10/3, 7/2] & -833/4096 + 17/81 = 2159/331776 \\ [7/2, 4] & -1 + 17/16 = 1/16. \end{array}$$

All are positive. When $t = 0$, both sides equal one. $\square$

For $F = \{0, 2, 7, 8, 11\}$, all 15 unordered pair sums are distinct, and direct addition gives

$$4F = \{0, 2, 4\} \cup \{6, 7, \dots, 38\} \cup \{40, 41, 44\}. \quad (11.3)$$

Thus $|4F| = 39$ and Lemma 11.4 applies.

11.3 Rational certificates on the weight orders

It remains to compare both U_c and V_c with $T_F(k)^{1/4}$. Use the rational majorant

$$\bar{V} = aK + b(S - K), \quad a = \frac{41}{80}, \quad b = \frac{7027}{14400} = \frac{c - a}{4}.$$

Indeed $a + 1/a = 8081/3280 < c$, so $\alpha < a$. The function $uK + (c - u)(S - K)/4$ has derivative $(5K - S)/4 \geq 0$; therefore $V_c \leq \bar{V}$.

Index the five positions of F in increasing order. On a cone where their weights have order π , write

$$k_{\pi(j)} = m + d_1 + \dots + d_{j-1} \quad (1 \leq j \leq 5), \quad x = (m, d_1, d_2, d_3, d_4) \geq 0.$$

There are $5! = 120$ orders, with ties allowed. The two linear targets $U_c = q^U \cdot x$ and $\bar{V} = q^V \cdot x$ have vectors

$$\begin{aligned} q^U &= (1109, 900, 675, 450, 225)/450, \\ q^V &= (35488, 28461, 21434, 14407, 7380)/14400. \end{aligned}$$

Let $\mathcal{M} = \{\mu \in \mathbb{N}_0^5 : |\mu| = 4\}$ be the 70 unordered fourfold multisets. For $\mu \in \mathcal{M}$, put

$$z(\mu) = \sum_i p_i \mu_i, \quad W_\mu(k) = \prod_i k_i^{\mu_i}, \quad \mathcal{M}_z = \{\mu : z(\mu) = z\}.$$

There are 39 nonempty fibers $\mathcal{M}_z$ by (11.3). These witness monomials have no ordered-representation multiplicity: the operation is max-convolution.

For each order π and each target q , the certificate supplies 70 nonnegative integers n_μ , with common denominator $D = 10^9$, satisfying

$$\sum_{\mu \in \mathcal{M}_z} n_\mu = D \quad (z \in 4F), \quad \sum_{\mu} \frac{n_\mu}{D} W_\mu(k(x)) - (q \cdot x)^4 \text{ has nonnegative coefficients.} \quad (11.4)$$

The supplied verifier checks these conditions for all 240 pairs (π, q) . Its arithmetic uses integers throughout each coefficient comparison. If $q = v/Q$, the comparison for a degree-four exponent γ is

$$Q^4 \sum_{\mu} n_\mu [x^\gamma] W_\mu(k(x)) - D [x^\gamma] (v \cdot x)^4 \geq 0.$$

The program reconstructs every witness coefficient by enumerating the variable choices in its four slots, and expands the target by repeated polynomial multiplication. It checks the

fiber masses, all 70 coefficients, and both targets for every order. The coefficient inequalities are checked independently of the solver’s floating-point values.

For every $x \geq 0$, (11.4) implies

$$T_F(k) = \sum_z \max_{\mu \in \mathcal{M}_z} W_\mu(k) \geq \sum_\mu \frac{n_\mu}{D} W_\mu(k) \geq (q \cdot x)^4.$$

Every witness has m^4 coefficient one. Fiber normalization therefore makes that coefficient 39 in the middle polynomial, whereas both target fourth powers have coefficient $c^4 < 3721/100 < 39$. Thus both comparisons are strict when $m > 0$.

Proof of Theorem 11.1. For positive k, g on F and nonzero f , combine the allocation envelope with the two certificate comparisons and the corner lemma. Each of U_c, V_c, C is strictly below $T_F(k)^{1/4}$, so the reflected inequality is strict.

For fixed f , all relevant norms are finite sums of maxima of finitely many coordinate products, and hence continuous in k, g . Positive approximation gives the non-strict inequality when some coordinates vanish. When $k > 0$, the upper envelope depends only on k and stays strictly below $T_F(k)^{1/4}$; taking the same limit in the envelope therefore preserves strictness for nonzero g supported on F .

For general g , restrict it to $g_F = g|_F$. The numerator is unchanged, while $\|f \star g_F\|_1 \leq \|f \star g\|_1$. If $g_F \neq 0$, apply the preceding result. If $g_F = 0$, the numerator is zero; when $k > 0$ and $f, g \neq 0$, the right side is positive. The zero-function cases of the non-strict statement are immediate. $\square$

The certificate files and standard-library checker accompany the paper. From this paper’s source directory, run

```
python3 -B verify_five_point.py
```

The command checks all 240 certificates and corruption controls. It requires no numerical optimizer or external Python package. The finite certificates establish the coefficient comparisons; the preceding hand argument supplies their connection to the reflected inequality.

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